INSTITUT DE RECHERCHE EN SCIENCES PSYCHOLOGIQUES (IPSY)

APPLICATION TO THE ETHICS COMMISSION FOR APPROVAL OF A RESEARCH PROJECT INVOLVING THE PARTICIPATION OF HUMAN BEINGS

Your opinion and suggestions regarding this document are important for the Ethics Commission. You can send us your comments by e-mail to demande-ethique-ipsy@uclouvain.be. If you wish to do so in an anonymous way, you can also send an unsigned letter to the president of the ethics commission: Nicolas Vermeulen, UCL-Institut de recherche en sciences psychologiques, Place du Cardinal Mercier, 10, 1348 Louvain-la-Neuve.

PART A. Project identification

1. **Name of the principal investigator**[1]: Pranav Pandya

   Title of the experiment: Assessing soft mobility comfort - walking and biking across distinct urban environments

For your information:

The signatures certify that the information contained in this form has been given in good faith and to the best of your knowledge.

You hereby certify that the study(ies) conducted in the framework of your project will be carried out in accordance with the ethics code of your profession, and in strict compliance with the physical and moral integrity of the participants.
As an investigator, your signature at the bottom of this document attests that you will assume the role and responsibilities that are assigned to you during the execution of this project. You also undertake to obtain prior authorisation from the Ethics Commission before carrying out any substantial change to this project.
demande-ethique-ipsy@uclouvain.be
If a mémorant or researcher sends a request for advice to the ethics commission, it is imperative that the sponsor be copied.

Date: 28/May/2025

Investigator's signature (permanent researcher in charge of the study, i.e. promoter):



Please tick the following box:
☒ **As the person responsible for the study, I undertake that all persons involved in the study will respect the ethical protocols, including confidentiality and anonymity.**

[1] Please note that for a thesis project, it is mandatory that the project is signed by the doctoral student and the promoter.

As the research supervisor, your signature at the bottom of this document attests to the fact that you will assume the role and responsibilities of the supervision of this project.

Date: 28/May/2025

Martin EDWARDS

Supervisor's signature (if applicable):



2. Purpose of the research

☒ PhD Thesis

Starting date of thesis: Academic Year: 2022-2026

☐ Academic research ☐ Sponsored research – Specify:

Starting date of research: 9/6/2025 End date of research (estimated): 9/10/2025

Any further details:

☐ Master Thesis

3. Principal investigator[2]

First and Last name: Pranav Pandya

Status: *PhD Student*

Specify the discipline if you are a PhD student: Art de bâtir et urbanisme

Institute/Center/Department: Louvain Research Institute for Landscape, Architecture, Built Environment (LAB)

Phone number: +32 499 35 11 35

E-mail: Pranav.pandya@uclouvain.be

4. Academic or scientific supervisor

First and Last name : Martin Edwards

Institute/Center/Department : IPSY

Phone number : +32 1047 87 88

E-mail : Martin.edwards@uclouvain.be

5. The institution(s) where the research will be conducted

Name: Urban enviornment surrounding LOCI - Uclouvain Brussels

Postal address: Rue Henri Wafelaerts 47/51, 1060 Saint-Gilles, Brussels

If the research is not conducted in a specific institution, specify the modalities (e.g. volunteers' homes, survey in railway stations, ...):

6. Discipline of the study

Other discipline

If other discipline, specify : Environmental Psychology

If you have obtained an agreement after completing this application, please keep a copy for future reference.

[2] The principal investigator is usually the person in charge of the actual execution of the project, e.g. the doctoral student; in the case of a PhD thesis, the approval of the principal promoter is required and point 4 should be completed.

Please answer all the questions on this form. If the question is not relevant to your project, please indicate N/A. At the end of your work, please save the file in .pdf format for dissemination within the committee.

PART B. Programme description: Scientific information

IMPORTANT INSTRUCTIONS

Please do not print and scan sections B, C and D of the form.

In order to avoid scanning part A, the e-mail message mentioning the approval of the promoter is also acceptable.

Please submit all documents in one PDF file. If you do not send all your documents in one PDF file, your application may not be considered. Other formats (.doc, .jpeg, etc) will not be accepted.

Save each document as a PDF and merge different PDF documents into one using software available for example here.

☒ **For approval**

1. Summary of the research programme (1 page); present a summary of the programme in a language that is understandable to all psychological researchers, including the following points.

1.1. Aim of the investigation

Title: Assessing urban mobility comfort - commuting across distinct urban environments
Have you already submitted your project to another ethics commission? Yes☐ No☒

If YES, which one? Cliquez ou appuyez ici pour entrer du texte.

If YES, what was the commission's decision? Cliquez ou appuyez ici pour entrer du texte.

My project is related to clinical psychology (or clinical orthopedagogy): Yes☐ No☐

To read the definition of clinical psychology, click on the following link :
https://uclouvain.be/fr/instituts-recherche/ipsy/procedures.html

1.2. Theoretical framework (about 5 lines)

Multi-sensory discomfort arises from complex environmental exposures, yet most studies focus on stationary subjects (1, 2). This overlooks the dynamic interaction between environment, perception, and stress in mobile users (3, 4, 5, 6). Thermal acclimatization—how quickly the body adjusts between shaded parks, exposed urban streets, or air-conditioned transport—takes up to 25 minutes indoors (7), but remains unclear outdoors.

1.3. Aims, hypotheses, research questions, contributions to knowledge development (approx. 5 lines)

This study explores commuter comfort via walking, cycling, and tram through different urban environments. Hypothesis: 1) Participants will show improved physiological and psychological well-being in green vs. impervious areas. 2) Acclimatization time will be estimated for green and impervious spaces. 3) Physiological and psychological responses will align. 4) Noise & activity will amplify thermal discomfort.

1.4. Description of the population of interest and of the concrete **procedures** used, in a way that makes it clear what is expected of the participants (approx. 20 lines)

1. Population: This study will recruit 30 participants (aged 18–40) with equal gender representation (50% male, 50% female), determined via power analysis (in Appendix A).

2. Experiment: Participants will engage in three phases: 1) a green environment, 2) an impervious urban environment, and 3) a tram journey. In the green and impervious phases, both walking and biking (tricycle) will be performed for 30 minutes each; during the tram phase, participants will remain seated for 30 minutes. In each phase there will be a 10-minute break for participants to rest, drinking water will be provided. Duration of experiment 1:30–5:30 pm in Etterbeek, with tram used to transition between phase 1 & 2.

3. Participants will wear four non-invasive sensing devices: 1) Empatica Embrace Plus smartwatch (measuring EDA, PPG, wrist skin temperature), 2) Pupil Labs Neon smart glasses (eye-tracking), 3) LYS Button (measuring illuminance) and 4) Atmotube PRO measuring air pollution. Sensors 3 & 4 will be worn on participant's shirt or jacket.

4. Environmental conditions will be measured by a researcher carrying a uCM backpack system (thermal stress, temperature, humidity, radiation, wind, sound, GPS, comfort indices) (8). If any clarification is required for environmental sensors, head of ethics commission of LAB – Prof. Sergio Altomonte has proposed to get in touch with him.

5. Participants will complete shortened psychological & comfort-related questionnaires (STAI-6 (10), PRS-11 (9,11), CSV, TSV (13)) & two custom questions before & after each exposure. Breaks are scheduled between sessions for recovery, survey completion, and sensor checks. Participation is voluntary, with a right to pause or withdraw at any time.

2. Methodological information	
a) **Data collection strategy:** Tick the procedure(s) used in your project	
A. Use of archival documents, files or data bases containing personal information not accessible to the general public	☐
B. Observation of individuals (several answers possible)	
B.1: direct observation by researcher visible to the participant	☒
B.2: direct observation by researcher hidden from the participant	☐
B.3: recorded observation that is known to the participant: audio	☐
recorded observation that is known to the participant: video	☐
B.4: recorded observation that is unknown to the participant	☐
C. Administration of a questionnaire	☒
D. Conducting individual or group interviews	☐
E. Performance of a task or administration of a test allowing the collection of psychological, psychometric, physical, intellectual or other measures	☐

F. Administration of an experimental medical treatment	☐
G. Collection of biological material	☐
H. Administration of products or substances	☐
I. Use of biological material originally obtained for medical purposes or from previous research or from outside researchers	☐
J. Physiological, neurological or electrophysiological investigation (EEG, MEG, MRI, PET, skin conductance, etc.). If you have ticked this point J, specify the technique(s) used and their possible invasive nature: As mentioned in Section 1.4, participants will be monitored only using non-invasive and non-intrusive wearable sensors. Empatica wristband is a smartwatch measuring heart rate, skin temperature and electrodermal activity. PupilLabs eye-trackers measure, visual fixation patterns, saccade movements (velocity, amplitude, direction), Pupil dilation response to environmental stimuli and track Areas of Interest (AOI). Additionally, participants will also wear LYS Button and Atmotube Pro on top of their shirt or jacket for measuring illuminance due to sunlight, and air pollution. Emotiv headset has 32 channel EEG sensor which can measure brain activity externally, EEG can be included to collect additional data if it is accepted by this commission.	☒
K. Other (specify) :	☐

b) Number of participants in the study

- Indicate the expected number of participants and briefly justify this choice of sample: 30
- Indicate if they are individuals / groups / families (specify): Individuals

c) Selection criteria for participants

- Study inclusion criteria: The participants must be aged between 18 and 40 years, residing in Brussels. They should be able to sit, walk, ride a bike and communicate in either French or English.
- Study exclusion criteria[3] : Individuals that are not in the age group 18 to 40 years will not be included. Individuals with impairment in walking and biking will not be included. In case that interested participants are wearing eyeglasses they should be able to replace them with contact lenses for the duration of the experiment so they can be equipped with eye trackers that are very similar to normal eyeglasses accounting for their comfort, safety and legibility of the data recorded.

d) **Recruitment of participants**

[3] This point does not concern data exclusion during the analysis, which can be justified for several reasons.

Indicate how you will obtain participants' contact details and how you will contact them. Attach in Annex D a copy of the announcement text or email that will be used or the text identifying the information that will be given during contact by phone, if applicable:

e) Recruitment period

Starting date of recruitment: Upon Approval (Estimated) end date of recruitment: June 2025

Comments, if any:

PART C. Ethical considerations

1. Have you requested the approval of a biomedical ethics committee for your study?

Yes ☐ No ☒

If yes, what was the opinion of this committee? (attach or specify if pending)

If no, and you are using one or more of the procedures mentioned in points F to J in the methodology used, or your project is related to clinical psychology*, please give reasons why you consider that the opinion of this committee is not necessary. * ***To read the definition of clinical psychology, click on the following link :*** *https://uclouvain.be/fr/instituts-recherche/ipsy/procedures.html*

This study evaluates commuter's muti-sensory comfort under distinct environmental conditions using only non-invasive and non-intrusive sensors along with environmental sensors. The study hypothesis are related to environmental psychology, and there is no potential harm to participants as per the literature (as mentioned below in sub-section 3 of this section.

2. Protections for vulnerable populations

a) **Minors, disabled adults, prisoners**

Does the research include:

- ***Minors*** : Yes ☐ No ☒
- ***Adults whose incapacity has been established by a court decision***: Yes ☐ No ☒

If you tick this box, you must obtain the consent of the persons who are legally responsible. You must also obtain the assent of minors or persons who lack capacity to understand the purpose and nature of the research.

Please explain how you will achieve this:

- ***Individuals in prisons***: Yes ☐ No ☒

b) **Dependency**

Does the research include:

- Individuals who are in a ***client-professional relationship*** with you, your supervisor or one of their associates? Yes ☐ No ☒
- Individuals who are in a ***student-teacher relationship*** with you? Yes ☐ No ☒
- Individuals who are in an ***employee-employer relationship*** with you? Yes ☐ No ☒

If you answered 'yes' to any of these questions, what strategy(ies) do you intend to use to preserve the freedom of these people to participate or not?

3. Risks and benefits (other than financial or material) for participants

a) What direct or indirect benefits can participants expect to receive from their participation?
Participants will have the opportunity to engage with wearable non-invasive and non-intrusive physiological sensing equipment, including heart rate, skin conductance, EEG (if it is accepted by this commission) and eye-tracking devices, and gain knowledge about how different urban environments influence comfort, stress, and emotional well-being.

b) What are the potential discomforts related to this participation such as, for example, negative affects (even temporary), deterioration of self-image (even temporary), worries or ruminations, physical discomfort (even slight), tiredness, or any other discomfort?

Please be aware that the vast majority of research in psychology and/or educational science leads to such inconveniences.[4]

Specify: Physical discomfort from wearing measurement devices (eye-tracking glasses, wrist sensors), fatigue from walking/biking.

What precautions are being taken to address these inconveniences?

• Clear participant briefing about all procedures and equipment before beginning

• Regular check-ins during the experiment with opportunities to rest

• Immediate removal of equipment upon request

• Post-session debriefing to address any discomfort or concerns

• Researcher accompaniment at a discreet distance to ensure safety

• A clean hand towel will be provided upon request to ensure participant comfort and hygiene

[4] This is particularly the case for studies
(1) using questionnaires measuring symptoms (e.g. depression, anxiety) or potentially dysfunctional psychological processes (e.g. emotional regulation strategies);
(2) using potentially difficult (e.g. tasks assessing executive, memory or attentional functions) or boring (e.g., tasks lasting longer than 1-5 minutes) laboratory tasks;
(3) presenting of stimuli likely to trigger emotional and/or psychological activation;
(4) using constraining methods (e.g. eye tracking, EEG, fMRI, TDCS);
(5) where the true purpose is not known to the participants (studies using experimental conditions in which participants are purposely deceived).

c) Does your research involve risks for participants that go beyond those encountered in everyday life: physical, psychological, social, professional, economic, political or other risks? Yes ☐ No ☒

If yes, specify:

If so, what precautions are planned to minimise known or anticipated risks to participants?

d) Identify, if applicable, the persons, services or organisations that can assist the participants:

4. Consent of the participants

Do participants give informed consent for their participation in the study?

☒ ***Yes, in written form*** (please provide a copy of this document in Annex D)

☐ ***Yes, in oral form***. In the case of oral consent, specify how it will be obtained and why written consent cannot be obtained:

☐ ***No***. If no consent is given, please explain the reasons:

Do participants give written consent for the use of their anonymised personal data (e.g. audio or video recording, photo, etc.)?

☒ ***Yes***, by the researcher only, ***for the analysis of the results***

☐ ***No***. If no consent is given, please explain the reasons:

For a minor or adult who is not capable of giving consent, describe the procedure for obtaining the participant's consent: *Click here to choose the appropriate procedure*

If other, specify:

If you found it difficult to choose, or could not give an answer for any of the items under this heading, please specify the reasons:

5. Partial disclosure and deception

In psychological research, it is generally accepted that the hypothesis being tested should not be disclosed to participants, in order to reduce effects of the investigator's expectations. The information provided to participants to obtain their consent is often general, with details being made available upon request at the end of the study. However, in some cases, the investigator is led to conceal the real purpose of the research or to claim a different purpose from the one being pursued. Such deception may cause psychological harm, especially if the misinformation relates to personal characteristics of the participants (personality, skills, dispositions, etc.). The following two questions aim to assess these risks.

a) Is there any information about the purpose or methodology of your study that should be hidden from participants for research purposes? Yes ☐ No ☒

If yes, Do you provide participants the opportunity to obtain more specific information about the research? Specify:

b) Does your research procedure involve temporarily misleading participants about the aims and procedures of this study? Yes ☐ No ☒

If yes, describe the nature of the subterfuge, and the reasons for using subterfuge:

If you answered yes to either of these two questions, please specify how and when you will inform the participants (method of desensitization – debriefing – offered to participants in relation to the subterfuge used):

6. Data confidentiality

Regarding data confidentiality, a distinction must be made between personal archives used for research purposes (videos, tests/retests, etc.) and information made public (courses, scientific communication).

a) Does the information stored allow the participant to be identified (e.g. name, date of birth, voice, photo, video image, etc.)? Yes ☐ No ☒

If yes, which information and for what purpose?

b) What procedures are in place to ensure the anonymity of participants and the confidentiality of data during processing and dissemination (e.g. coded transcription, data encryption or digitalisation, erasure of tapes, aggregated data, omission of certain characteristics, use of a pseudonym, destruction of questionnaires, etc.)? The data collected will be pseudonymized. Each participant will be assigned a code (initials of the experimenter - control group/experimental group - participant number) (e.g. PP - ctrl - 001 or PP - expe - 002). Each participant will be recorded as code and participants will never be identified by their first name or in any other recognizable way, in any lists, results or publications related to the study. These data will be stored on the principal investigator and co-investigators' personal computers, where only the persons concerned will have access to the data through a secured common cloud storage server operated by IT department to minimize the risk of unauthorized acces (e.g. Microsoft OneDrive and Uclouvain private server). All computers used by the project team will be password encrypted. The protection of personal data is ensured by the law of July 30, 2018, on the protection of privacy and by European (European General Regulation on the Protection of Personal Data [RGPD] of May 25, 2018) and Belgian regulations in force. According to the RGPD, participants have a right to inspect the processing of their data. If they have any questions in this regard, they may contact the promoter or UCLouvain's data protection officer, as explained in the consent letter. If they have a complaint about the way their data is processed, they can contact the Belgian Data Protection Authority: Rue de la Presse 35 - 1000 Brussels - Tel: 02 274 48 00 - e-mail: contact@apd-gba.be.

c) Will the results of the research be communicated to the research partner organisations in a form other than a publication or a general presentation of the results?

Yes ☐ No ☒ NA ☐

If yes, describe how the results will be transmitted and how you will ensure that the presentation of the results will preserve the anonymity of the participants:

d) In certain circumstances, you may be required by law to disclose certain information to third parties (e.g., reporting abuse). Have you included a clause to this effect in the consent form?

Yes ☐ No ☒ NA ☐

If the answer is yes and you are faced with such a situation, what strategy(ies) will you apply?

7. Compensation

a) Will participants receive compensation for their participation in the research?

Yes ☒ No ☐

If yes, in which modalities?

☐ a sum of money equivalent to the expenses incurred. Specify the amount and justify:

☒ a fixed sum of 15 euros

☐ a gift (voucher, lottery ticket, etc.). If applicable, please specify which one:

☐ credits for a course

☐ a psychological follow-up

☐ another form of compensation. Specify:

If no, justify:

b) Is the amount to be paid or the manner in which it will be paid the same for each participant?

☒ Yes

☐ No ***If no***, justify :

☐ NA

c) If a participant withdraws from the study, will they still receive compensation?

☒ In full

☐ Partly Specify:

☐ No Justify:

8. Feedback

At the end of the session, will you inform the participants of the objectives of the study and the means you used to achieve them ("debriefing")? For online studies, participants must be able to download the debriefing.

☒ Yes

If yes, Please provide in Annex D a list of the main points to be discussed during this debriefing.

☐ No

If no, justify:

Will participants have the opportunity to ask you questions and give feedback?

Yes ☒ No ☐ NA ☐

If no, justify:

At the end of the analyses, will the results of the research be communicated to the participants (individually or collectively)

- Individually: Yes ☐ No ☒

If yes, justify and draw participants' attention to the fact that these are preliminary results:

- Collectively: Yes ☒ No ☐

If yes, justify and draw participants' attention to the fact that these are preliminary results:

The committee recommends that individual results should not be reported. Caution should be exercised in the presentation of preliminary, non-peer-reviewed results.

9. Question to the Ethics Committee

Is there a particular problem or issue on which you would like to seek advice and/or assistance from the Ethics Committee?

PART D. Annexes

As a reminder:

Please submit all documents as a single PDF. Other formats (.doc, .jpeg, etc) will not be accepted.

Save each document as a PDF and merge different PDF documents into one using available software, e.g. here.

Documents to be attached (Max 3 Mo per folder)**:**

- Annex 1: The text of the informed consent for the participants and/or for the persons legally responsible for participants who are minors or declared incompetent;
- Annex 2: A copy of the correspondence confirming the authorisation of the institutions, organisations or establishments whose collaboration is necessary to carry out the research;
- Annex 3: A copy of the questionnaires, interview layouts or observation grids;
- Annex 4: A copy of the information that will be given for recruitment purposes: posters, advertisements, phone scripts or e-mails;
- Annex 5: For online studies, participants must be able to download the debriefing.

Annex 1

Participant Information Sheet

In this study, you will be asked to participate in a series of assessments during commuting routes in the Brussels Capital Region (BCR). The experiment will take place from 1:30 to 5:30 pm. In this time, you will alternate between walking and biking along urban bike and pedestrian routes. You will also be asked to take the tram. Researchers from UCLouvain will be accompanying you along these urban routes. You will wear non-invasive and non-intrusive wearable sensors to measure heart-rate variability, skin conductance, skin temperature, together with EEG (if it is accepted by ethics commission) and eye-tracking glasses and illuminance due to sunlight. You will also complete brief surveys about your emotional state and your perceptions of the urban environment. These surveys will be administered at the beginning & end of each phase before or after biking/walking, to track changes in your physiological responses and psychological state as you move through different urban settings.

Your participation is entirely voluntary. All information collected will be handled in accordance with GDPR and the Belgian Law of 30 July 2018 on the protection of individuals with regards to the processing of personal data. By signing this consent form, you agree to take part in this study and consent to anonymous statistical analysis of the data you provide. You may withdraw any or all of your data from the study at any time without consequence. All data will be stored securely under UCLouvain regulations; see https://www.uclouvain.be/en/privacy for details.

Informed Consent Form

- I have read the Participant Information Sheet, and the nature and purpose of the study have been explained to me.
- I understand the study's purpose and what my participation entails, and I agree to take part.
- I understand that I may withdraw from the study at any time, and this will not affect my status now or in the future.
- I understand that, while anonymized data gathered during the study may be published, I will not be identified in any form, and my personal information will remain strictly confidential.
- I understand that data from this study will be stored in a secure location in accordance with UCLouvain regulations, and only the investigator and supervisor will have access.
- I understand I may contact the investigator or supervisor at any time if I require further information about the study.

Please indicate your consent by signing your name and date below:

- **Investigator:** Pranav Pandya, pranav.pandya@uclouvain.be
- **Promoter:** Maider Llaguno-Munitxa, maider.llaguno@uclouvain.be
- **Co-promoter:** Martin Edwards, martin.edwards@uclouvain.be

Annex 2

The current study will be conducted only by this institution thus there is no additional documents to be attached in this Annex.

Annex 3

A) Background Information form

Background information forms will be filled at the beginning of the experiment (see table 1). These questions are designed to collect basic demographic, residential, and behavioral information that may help contextualize each participant's responses during the experiment. The form also captures participants' physical characteristics relevant to sensor use (e.g., vision correction and handedness), as well as their mobility habits and familiarity with the study area. This information will not be used to exclude participants but to aid in the interpretation of collected psychophysiological data.

Table 1: Participant Background Information

Category	Questions	Response Type
Identity	1. What is your name?	Text field
Demographics	2. What is your age?	Text field
	3. What is your gender?	Multiple choice: Male/ Female/ Prefer not to say
	4. What is your nationality or country of origin?	Text field
Residence	5. Where do you live in Brussels?	Text field
	6. How long have you been living in Brussels?	Text field
Physical Characteristics	7. What is your laterality? (Are you left or right-handed?)	Binary choice: Left / Right
	8. What is your current body weight? In Kg.	Text field
	9. What is your height? In cm.	Text field
	10. What are you currently wearing? Please describe your outfit as precisely as possible including the color (e.g., short-sleeved shirt, jeans, sandals, light jacket, etc.).	Text field
	11. What do you normally use for vision correction?	Multiple choice: Normal vision/ Wear glasses/ Wear contact lens / Uncorrected vision impairment/ Prefer not to say
Familiarity	12. How well do you know the streets and surroundings where the experiment is in Brussels?	Linear scale 1–10: Not at all familiar (1) / Extremely familiar (10)
Living Environment	13. What is your type of residence?	Binary choice: City / Suburban environment
Household	14. Do you have children?	Multiple choice: Yes / No
Mobility	15. How do you commute to work?	Multiple choice: Walk/ Bike/ Train/ Metro/ Tram
Physical Activity	16. Is there any sport or physical activity you practice regularly? Name it. Else write no.	Text field
	17. How frequently do you bike?	Multiple choice: Daily/ Few times per week/ Few times per month/ Rarely/ Never
	18. On average how much do you walk every day?	Multiple choice: <15 min/ 15 to 30 min/ 30 min to 1 hour/ 1 to 2 hours/ > 2 hours

B) Survey during the experimental campaign

These surveys include participants' thermal and psychological perceptual responses to urban environments during urban commuting. The survey incorporates elements from validated and shortened standardized questionnaires covering both thermal and psychological dimensions. These include the Comfort Sensation Vote (CSV) and Thermal Sensation Vote (TSV) for thermal perception (Xue et al., 2020), the 6-item State-Trait Anxiety Inventory (STAI-6) for psychological stress (Marteau and Bekker, 1992; Spielberger, 1983), and the 11-item Perceived Restorativeness Scale (PRS-11) for environmental restoration (Hartig et al., 1997; Pasini et al., 2014). Additionally, custom-made questions on comfort and environmental pleasantness (Environmental Stressor and Greenness) have also been included (see complete survey in table 2). The survey will require approximately 4 minutes of time to fill in.

As described in figure 1 the experiment will begin at 1:30 pm and will end at 5:30 pm. Upon arrival participants will be invited to sit in an air-conditioned room for the signing of the consent form, the installation of the physiological sensors, and the filling in of the 'participant background information' form and to complete the survey. Then the participants will be asked to either bike or walk along a predefined path for 30 minutes. Once this is completed, the sensors will be checked, and the participants will be asked to fill in the survey. Once this is completed another block of 30 minutes of either biking or walking will follow. Tricycle will be used for biking having 3 wheels which will prevent accidental fall and ensure safety. Once this is completed, once again the sensors will be checked, and the participant will be asked to fill in the survey. These two initial blocks of 30 minutes of biking and walking will take place in either green or impervious urban environments – choice that will be randomized as it will be the order on which the participant will walk or bike first. This initial block of data collection will be followed by a tram route with a total duration of ~30min where the participant will be sitting in an air-conditioned environment. This will bring the participant back to the initial location to start block two, which will be organized as block one. Before the following 30 minutes of biking or walking, the participant is asked to fill in the survey once more. Block two will unfold as per block one after which the participant will have completed the experiment. Along with the experimental campaign, refreshments will be provided.

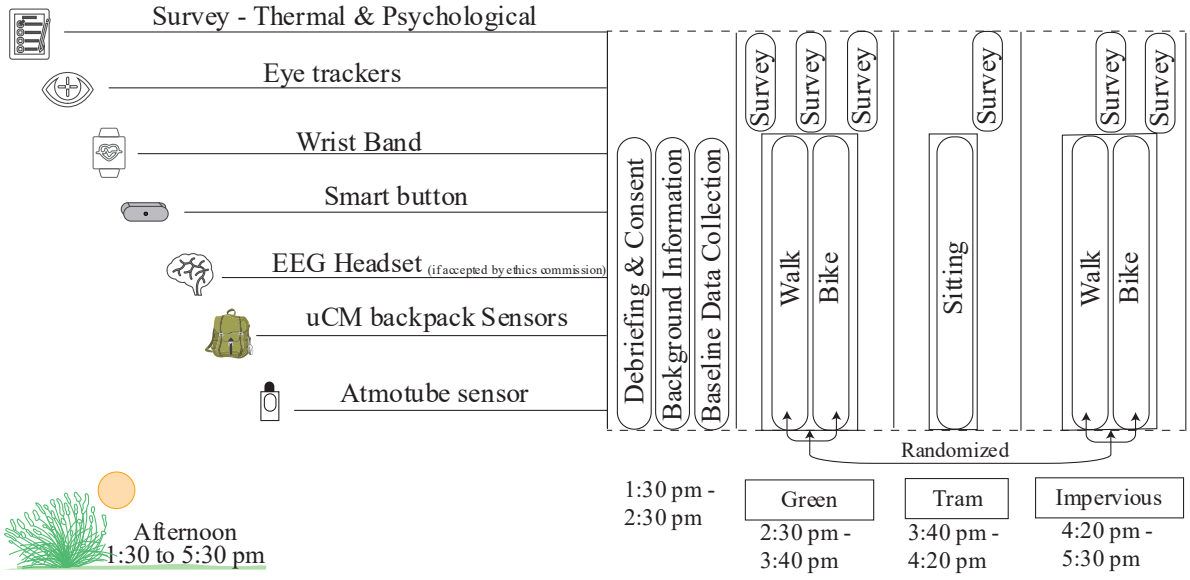


Figure 1. Experimental Campaign Timeline

Table 2: Experimental Campaign Questionnaires

Item	Question	Response Format	Source Framework
1	How comfortable do you feel right now?	Very uncomfortable / Uncomfortable / Slightly uncomfortable / Neutral / Slightly comfortable / Comfortable / Very comfortable	Comfort Sensation Vote
2	Please describe your thermal sensation	Cold / Cool / Slightly cool / Neutral / Slightly warm / Warm / Hot	Thermal Sensation Vote
3	Which environmental factor is most affecting you right now?	□ Temperature □ Lighting □ Noise □ Air Quality □ Crowding	Environmental Stressor
4	I feel calm	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
5	I feel tense	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
6	I feel upset	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
7	I feel relaxed	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
8	I feel content	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
9	I feel worried	Not at all / A little / Moderately / Quite a lot / Extremely	STAI
10	Places like that are fascinating	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
11	In places like this my attention is drawn to many interesting things	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
12	In places like this it is hard to be bored	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
13	Places like that are a refuge from nuisances	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
14	To get away from things that usually demand my attention I like to go to places like this	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
15	To stop thinking about the things that I must get done I like to go to places like this	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
16	There is a clear order in the physical arrangement of places like this	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
17	In places like this it is easy to see how things are organised	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
18	In places like this everything seems to have its proper place	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
19	That place is large enough to allow exploration in many directions	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
20	In places like that there are few boundaries to limit my possibility for moving about	Not at all / A little / Moderately / Quite a lot / Extremely	PRS
21	How pleasant do you feel in this place?	Not at all / A little / Moderately / Quite a lot / Extremely	Greenness

STAI (State-Trait Anxiety Inventory)

- Spielberger, C. D., Gorsuch, R. L., Lushene, R., Vagg, P. R., & Jacobs, G. A. (1983). Manual for the State-Trait Anxiety Inventory. Palo Alto, CA: Consulting Psychologists Press.
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PRS (Perceived Restorativeness Scale)

- Hartig, T., Korpela, K., Evans, G. W., & Gärling, T. (1997). A measure of restorative quality in environments. Scandinavian Housing and Planning Research, 14(4), 175-194.
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CSV (Comfort Sensation Vote) & TSV (Thermal Sensation Vote)

- Xue, J., Hu, X., Sani, S. N., Wu, Y., Li, X., Chai, L., & Lai, D. (2020). Outdoor Thermal Comfort at a University Campus: Studies from Personal and Long-Term Thermal History Perspectives. Sustainability, 12(21), Article 21. https://doi.org/10.3390/su12219284
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Annex 4

Recruitment Poster



Figure 2: Poster – Participation Call

Call to be shared for the experimental campaign:
We are recruiting 30 individuals aged 18-40 for a research study examining comfort, stress, and perception during walking and biking. The study aims to explore whether individuals feel greater discomfort in impervious areas, whether green spaces reduce stress and enhance emotional well-being, and how different modes of movement (such as biking versus walking) may vary in perceived effort or stress.

Participation involves performing walking, biking and riding trams in Brussels in green and urban spaces. You'll wear non-invasive and non-intrusive sensors measuring heart rate, skin conductance, and temperature, along with eye-tracking glasses and button for measuring illuminance due to sunlight, while completing brief surveys about your emotional state. All instruments are safe, your data will remain confidential, and you may pause or withdraw at any time. If interested, please contact us here: pranav.pandya@uclouvain.be.

Annex 5

Debriefing Participants

Thank you for taking part in the experiment "Assessing urban mobility comfort - commuting across distinct urban environments."

What We're Hoping to Learn:

i) You may have felt more discomfort in areas without greenery, especially when it was hot.
ii) Green spaces may help reduce stress and improve how people feel.
iii) Urban noise may have felt more discomforting with in thermal stress.
iv) We're also checking if what people report feeling matches their body's signals (like heart rate or sweating).

All the information you provided will remain confidential and will only be used for research purposes. You maintain the right to withdraw your data within 30 days by contacting us. If you have any questions about this research, please reach us out at pranav.pandya@uclouvain.be

Appendix A

Sample Size Calculation for the Experimental Campaign

Working Hypotheses

1- Participants will show improved physiological and psychological well-being in green vs. impervious areas.
2- Acclimatization takes longer after green and air-conditioned settings due to greater heat stress contrasts.
3- Physiological and psychological responses will align.
4- Noise & activity will amplify thermal discomfort.

Software Libraries

1. NumPy: Numeric arrays and basic calculations
2. SciPy: Statistical tests and curve fitting
3. Statsmodels: Power analysis routines (TTestPower, FTestAnovaPower)
4. Pingouin: Repeated-measures ANOVA power (power_rm_anova)

Statistical Methodology

Three a priori power analyses align with the primary within-subject design:

1. Paired t-test for the diurnal contrast (Green vs Impervious).
2. Repeated-Measures ANOVA across mobility modes (walk, bike, tram), each performed in afternoon.
3. Wilcoxon Signed-Rank Test for comparing subjective and physiological responses (e.g. reported vs. measured comfort).

Key Statistical Parameters

- Significance level (α): 0.1
- Target power ($1 - \beta$): 0.80
- Within-subject conditions: 5 (walk-green, bike-green, tram, walk-urban, bike-urban)
- Effect-size grid (expecting medium to large effect size):
    - Paired t: Cohen's d = 0.50 (Medium), 0.80 (Large)
- Cohen's f notation (medium = 0.25, large = 0.40)
- Dropout cushion: +5% recruitment

Table 3: Sample Size Analysis Results

Analysis Method	Effect Size Category	Effect Size Value	Required N	Notes
Paired t-test	Medium	d = 0.50	27 participants	Can detect medium differences between two conditions measured on the same participants (Hypothesis 1: green vs. impervious).
Paired t-test	Large	d = 0.80	12 participants	Can detect large differences between two conditions measured on the same participants (Hypothesis 1: green vs. impervious).
Repeated-Measures ANOVA	Medium	f = 0.25	17 participants	Can detect medium differences across several repeated conditions per participant (Hypothesis 2 & 4: interaction of mobility mode and environment type — walking, biking × green, impervious).

Repeated-Measures ANOVA	Large	f = 0.4	8 participants	Can detect large differences across several repeated conditions per participant (Hypothesis 2 & 4: interaction of mobility mode and environment type — walking, biking × green, impervious).
Wilcoxon Signed-Rank Test	Medium	d = 0.50	28 participants	For detecting medium changes between two related non-normally distributed measures (Hypothesis 3: subjective vs. physiological response).
Wilcoxon Signed-Rank Test	Large	d = 0.80	12 participants	For detecting large changes between two related non-normally distributed measures (Hypothesis 3: subjective vs. physiological response).

Conclusion

This study will recruit 30 participants. This sample size of 30 was obtained based on the following statistical calculations (with the assumption that medium effect size is achieved):

1. Paired t-tests are used to detect mean differences between two conditions measured within the same participant. In this experiment, they are applied to compare psychological and physiological outcomes between green and impervious urban environments (Hypothesis 1), as well as between biking and walking. Based on a medium effect size (d = 0.50), this results in a sample size requirement of 27 participants.
2. Repeated measures ANOVA can detect differences across several repeated conditions within participants. In this experiment, each participant is observed under five mobility–environment combinations (walking, biking, and tram, across green and impervious settings), allowing assessment of interaction effects such as acclimatization and thermal comfort dynamics (Hypothesis 2 & 4). Considering a medium effect size (f = 0.25), a required sample size of 17 participants was obtained.
3. Wilcoxon Signed-Rank Test is applied to assess the association between subjective thermal comfort responses and physiological sensor data, such as heart rate, EDA, or skin temperature (Hypothesis 3). With a medium effect size (d = 0.50), the sample size requirement is 28 participants.

Based on the more conservative requirement of 28 participants and accounting for a 5% potential attrition rate, the study will recruit 30 participants in total. This within-subject design allows each participant to experience all conditions, minimizing individual variability and fatigue, while enabling the collection of continuous real-world data and ensuring sufficient statistical power to test the primary hypotheses. Hypothesis 5, which examines the potential amplifying effect of urban noise and activity on perceived thermal discomfort, is considered exploratory and was not included in priori power calculations. It will be examined descriptively using environmental data (e.g., sound pressure levels), but no formal statistical testing is planned.

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Appendix B

Representation of Sensors which will be worn by participants

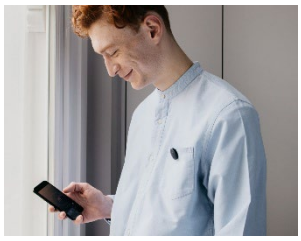
Participants will wear the following non-invasive and non-intrusive sensors as shown below:



Empatica Embrace Plus (Smart Watch)



Pupil Labs Neon smart glasses (eye-tracking)



LYS Button (measuring illuminance)



Atmotube PRO measuring air pollution



Emotiv 32 channel EEG headset (if it is accepted by this commission)

Appendix C

Time period selection for experiment

SOLWEIG (Solar and Long Wave Environmental Irradiance Geometry) model was used to simulate mean radiant temperature (T_{mrt}) for 2022 to 2024. Human thermal comfort indices - Physiologically Equivalent Temperature (PET) and Universal Thermal Climate Index (UTCI) were generated. These were calculated using the Pythermalcomfort library with inputs including Tmrt, wind speed, relative humidity, and air temperature.

On hourly plot, T_{mrt} peaked at roughly 53.9 °C around midday (when solar radiation is highest), but PET and UTCI—because they also respond to the slower-warming air mass and the cooling effects of wind—peaked later at 15 & 16 hour.

PET and UTCI values are elevated between 2:00 pm and 5:00 pm. To capture this peak period while allowing time for consent and preparation, participants are asked to arrive at 1:30 pm. Therefore, the experimental window was set from 1:30 pm to 5:30 pm.

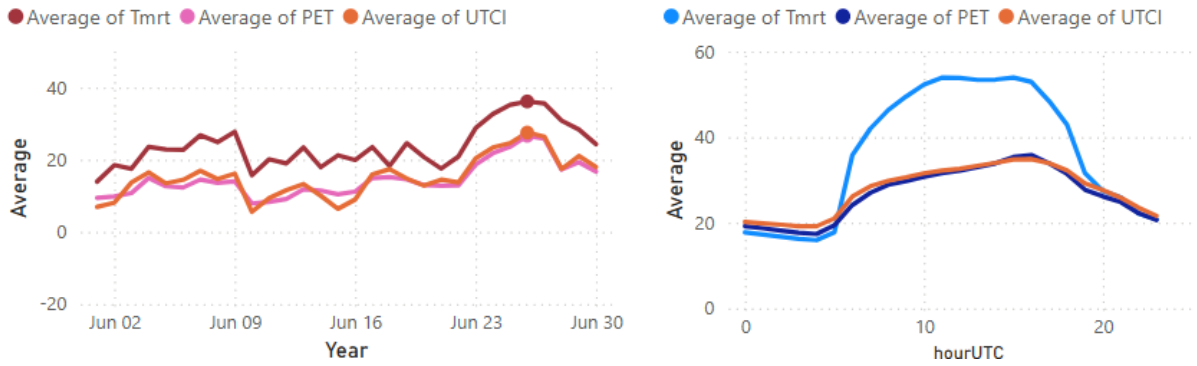


Figure 3: Plot of Tmrt and thermal comfort indices – PET & UTCI for impervious environment.

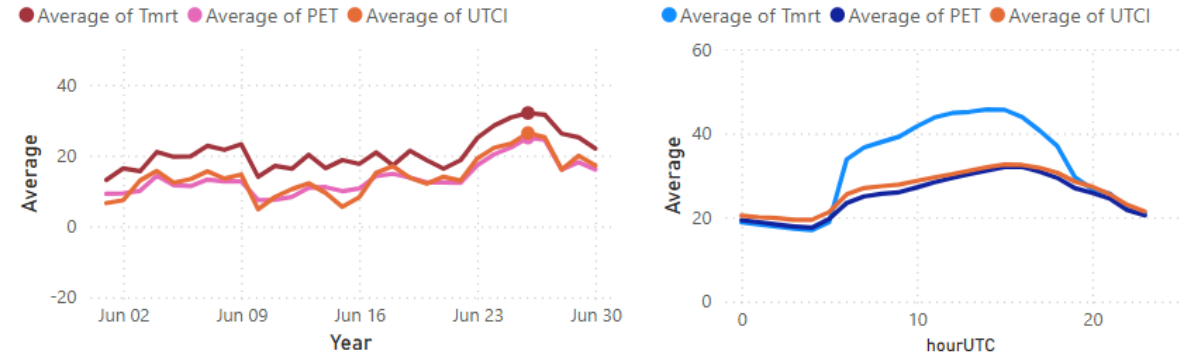


Figure 4: Plot of Tmrt and thermal comfort indices – PET & UTCI for green environment.

Appendix D

Route Safety Planning of Experiment

- The participants will always be accompanied by a PhD researcher.
- Tricycle will be used for biking having 3 wheels which will prevent accidental fall and ensure safety.
- The experiment will be done in dedicated bicycle paths which ensures that safe distance from participants will be maintained and there will be no risks such as overtaking or sudden stoppage.
- The dedicated bike paths have separate signals for bicycle riders which prevent the risk of accidents.
- In the planning of experiment there is a 10 minute of break after every two blocks (1 break in green and 1 break in urban areas) during which they must walk and ride a bike. This break time is always available to participants upon request.

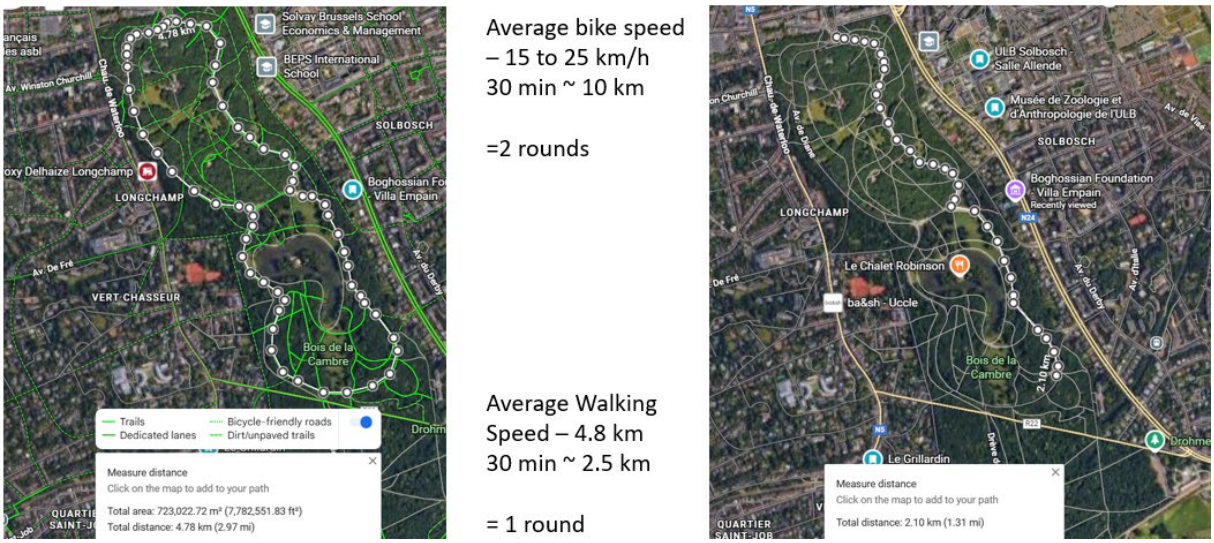


Figure 5: Green Environment Bicycle and walking path

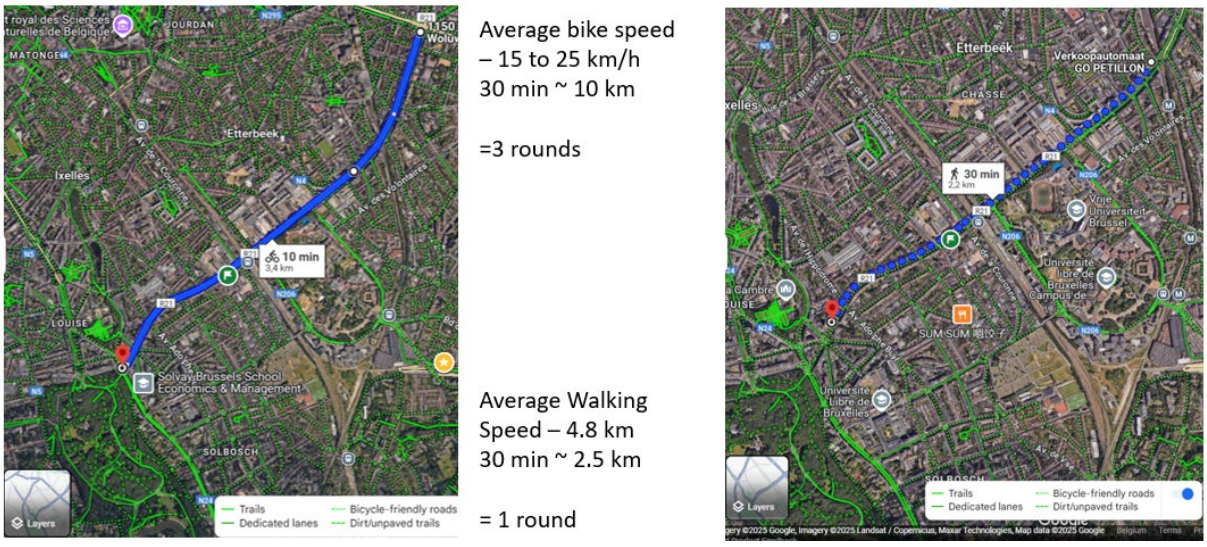


Figure 6: Impervious environment bicycle and walking path

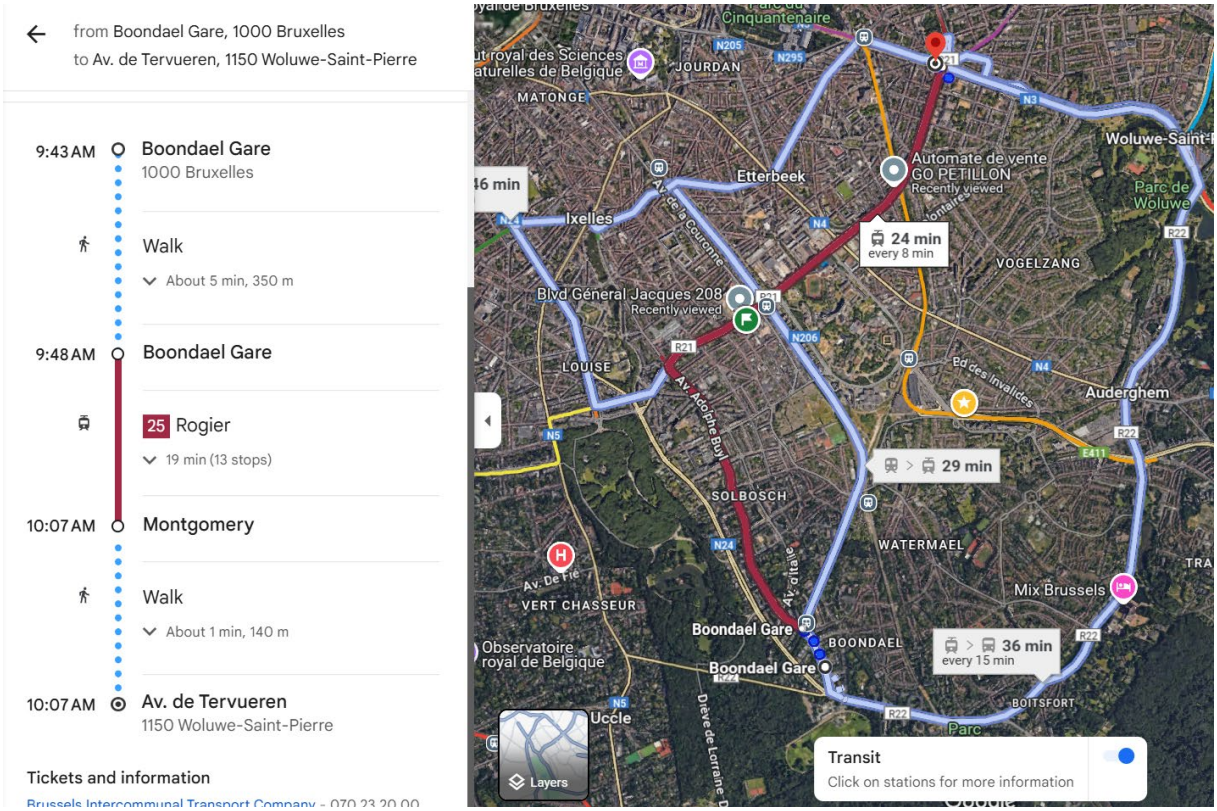


Figure 7: Tram journey

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How to Measure The Restorative Quality of Environments: The PRS-11

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Abstract

The Perceived Restorativeness Scale (PRS) has been reported relatively frequently in the literature, despite the psychometric and factorial properties of the scale not being well established. We argue that a detailed understanding of the meaning of individual items is the proper starting point for scale development and used this approach to develop shorter (11-item rather than 26-item) parallel versions in both Italian and English. Data collected from samples of Italian (n = 230) and English speakers (n = 100) were analysed by Confirmatory Factor Analysis (CFA), comparing a 5 models based on previous published research and underlying theory. A four-factor model that mirrored four elements of Attention Restoration Theory (ART) had the best fit to the data. The resulting composite scale was invariant across nationality and gender.

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Peer-review under responsibility of the Academic World Education and Research Center.
Keywords: Restorative environments, validation, PRS scale, multigroup CFA; cognitive interviewing ; ART

1. Introduction

Despite the fact that use of the Perceived Restorativeness Scale (PRS); (Hartig, Kaiser, & Bowler, 1997a) has been reported relatively frequently in the literature dealing with the extent to which particular environments have restorative qualities, our review of this literature suggests that the psychometric properties of the scale have not been established in a definitive way. In part, this is because some items vary from study to study and in part because the majority of studies are aimed at using rather than analyzing the scale (Bodin & Hartig, 2003; Lehto, 2013; Pals,

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doi:10.1016/j.sbspro.2014.12.375

Steg, Siero, & Van der Zee, 2009; Norling, Sibthorp, Ruddell, 2008). The PRS is based on the Attention Restoration Theory (ART; Kaplan, 1995) and it was initially made up of 26 items aimed to measure an individual's perception of 5 restorative factors assumed to be present to a greater or lesser extent in the environment. These factors are physical and/or psychological "being-away" from demands on directed attention; "fascination" a type of attention assumed to be effortless and without capacity limitations; the "coherence" and "scope" perceived in an environment; the "compatibility" between one's inclinations and environmental demands (for more details see Berto, 2005; Purcell, Peron, Berto, 2001). As originally formulated, ART focussed on 4 restorative factors: being-away, fascination, extent and compatibility. However, extent was subsequently thought to comprise two elements: the "coherence" and the "scope" of an environment. Since Hartig et al.'s work (1997a) there have been only two published studies aimed at assessing the psychometric characteristics of the scale (Pasini Berto, Scopelliti, & Carrus, 2009; Han, 2003). Given the differences in the items across studies, it is difficult to match PRS factor solutions to each other and in a precise way to ART. The articles describing the initial development of the PRS (Hartig et al., 1997a; Hartig, Korpela, Evans, Gärling, 1997b) do not provide a detailed rationale for the item development and, hence, the likely validity of any particular item.

In the research described here, we set out to do two things. The first was to develop a shorter version of the PRS to make it more suitable for use in research contexts where time is limited, and the second was to establish the psychometric properties of the items and overall scale in a more robust way. We began the review of each item in the original scale using cognitive interviewing with Italian, Canadian and Australian subjects to gain a better understanding of the language used by laypeople when discussing the restorative quality of environments. This process also gave us a better understanding of how people interpreted the existing wording of the items and the link between the items and the theoretical a-priori factors (ART; Kaplan, 1995). Developing a scale that can be used in multiple languages presents a number of challenges. For example, differences in the verb tenses between the Italian and English versions of the 26-item PRS affected the interpretation of "being away". The present tense produced the greatest degree of clarity in people's understanding of the items and this tense was adopted for the development of the shorter scale in both languages. In line with previous research (Hinkin, 1995; Pals et al., 2009), our interviews showed it to be important to avoid double negation and to use positive wording.

A comparison of the high Alpha value for the 26-item PRS (0.94, Berto, 2007: 0.95; Pasini et al., 2009) with the Alpha for a scale with only 5 items (0.79; Berto, 2005) suggested that the alpha measure may not be telling us anything very helpful. This suggested that not only the number of items matters but also their meaning needed review.

Even the PRS's authors struggled to settle on (i) the number of items making up the scale and (ii) the factorial structure (Hartig et al., 1996). However, across published studies, two components of a restorative experience are consistently identified present in each analysis: "being-away" and "fascination" (Pasini & Berto, 2007; Pasini et al., 2009; Purcell et al., 2001). These are the core features of the ART (Kaplan, 1995; Kaplan & Kaplan, 1981).

From our exploratory analysis and from our review of the literature it turns out that the "compatibility" items have not loaded onto a separate factor in any convincing way (Hartig et al., 1997a; Pals et al., 2009; Pasini & Berto, 2007). This may be due to a conceptual problem in that "compatibility" is a loose kind of personality measure embedded in the ART. Consequently, we did not include the compatibility items when developing the shorter scale being described here.

2. Method

2.1. Participants

This study involved 330 participants, 230 from Italy and 100 from Canada. Mean age was 23.5 years (from 18 to 45, sd 6.08), and 55.7% were females.

2.2. Material

Ten colour photographs of outdoor environments were used. The photographs spanned the range from totally natural to totally built environments as well as the entire range from low to high preference and restorativeness scores

(Berto, 2005, 2007; Hernandez et al., 2001; Purcell et al., 2001). The photographs represented five environmental categories (two photographs per category): Industrial zone, housing, city streets, hills, lakes.

2.3. Measurement instrument

Seventeen items were selected from the original 26-item version of Hartig et al.'s (1997a) and Pasini et al.'s (2009) instruments, and in some cases modified, using the cognitive interview technique (Willis, 2005) and recommendations from previous research. The original 6 being-away items were all preserved, while one of the 7 fascination items was considered to be tapping the scope dimension (Tenngart & Hagerhall, 2008). Finally, two scope items; one coherence item and the 6 compatibility items were removed. Both an Italian and an English translation were drafted by native speakers and checked by reverse translation. Subject's judgments were made on a 0 to 10-point scale, where 0 = not at all, 6 = rather much, and 10 = completely.

2.4. Procedure

Participants were randomly assigned to 10 subgroups. One of the 10 photographs was randomly assigned to each subgroup. Subjects received a copy of the PRS-17 and a sheet with the following instructions: *We are interested in how you experience this environment. To help us understand your experience, we have provided the following statements for you to respond to. Please read carefully, then ask yourself: "how much does this statements apply to my experience there?". To indicate your answer, circle only one numbers on the rating scale beside the statement. A sample of the rating scale is given below and at the top of each subsequent page. So, for example, if you think that the statement does not at all apply to your experience of the environment, then you would circle "0" (not at all), if you think it applies rather much, then you would circle "6" (rather much), but if you think that it apply very much, you would circle 10 (very much)*. If the subject correctly understood the instruction then the photo was shown. Subjects had to rate each item of the PRS-17 in relation to the photograph.

2.5. Data analyses

Confirmatory Factor Analysis (CFA) was used to find the best factor structure of the scale, using both the ART concepts and previous factorial studies to inform the analysis. In a second step, the best model was tested for measurement invariance across pictures, gender and country. Finally, ANOVA was used to check differences in the ratings across the different environmental categories.

3. Results

Five models were compared (see Table 1) using CFA. At first the model with 17 items predicted by 3 latent factors (Being away, Fascination and Extent) was estimated (Model 1). It should be noted that the Fascination item 10 in Hartig et al.'s scale (1997a) ("*Places like that are large enough to allow exploration in many directions*"), also included in this 17-item version, has been interpreted as a Scope item following Tenngart & Hagerhall's suggestion (2008). Model 1 showed bad fit indexes and some very low factor loadings (e.g. .39). Therefore, we decided to delete some items from the scale. This process was guided by previous findings in the restorativeness literature and by the examination of the item statistics and modification indexes from the CFA analysis. We chose to maintain items common to previous research and to eliminate items with low factor loadings, and also items that have been found to be problematic in previous studies. At the end of this process, 11 items remained, and some alternative models were estimated to compare with some of the more convincing factor structures reported in the literature. First, we analysed a model (Model 2) with 11 items predicted by a general factor. Factor loadings ranged from .25 to .83 and fit indexes were unacceptable. A third estimated model (Model 3) followed suggestions by Herzog (2003) and Tenngart & Hagerhall (2008), and was composed of 3 factors: Being away (3 items), Fascination (3 items) and Extent (5 items). Model 3 showed a great improvement in all fit indexes, but some factor loadings remained problematic (e.g. .34). We decided to estimate models that assumed the two components of Extent (Coherence and Scope) as distinct. Therefore, the fourth model (Model 4) was composed by 3 factors (Being away, Fascination and

Coherence) following Hartig and Purcell's suggestions (Hartig et al, 1997a; Purcell et al., 2001). Each factor predicted 3 items. Fit indexes were excellent and factor loadings ranged from .43 to .88. The last model (Model 5) added the two-item Scope dimension. Fit indexes were good, even if worse than Model 4, but showed, on average, higher factor loadings than Model 4. On the basis of its statistical characteristics and the fact that a four dimensional model might provide a more complete picture of restorativeness, we chose this final model to use for further analysis (see Appendix A for the PRS-11 final version).

Table 1. Fit indexes and comparisons between the alternative models; BA =being-away, FA = fascination; EXT = extent; COH = coherence; SCO = scope

Competitive models	χ^2 (df)	$\Delta\chi^2$ (df)	CFI	TLI	RMSEA	SRMR	AIC	BIC
Model 1. Four factors (BA, FA, EXT) – 17 items	528.51(116)		.85	.83	.10	.07	602.51	743.07
Model 2. General factor – 11 items	267.71 (44)	260.80 (72)***	.86	.82	.12	.07	311.71	395.29
Model 3. Three factors model (BA, FA and EXT) – 11 items	146.09(40)	121.61 (3)***	.93	.91	.09	.05	198.09	296.87
Model 4. Three factors model (BA, FA and COH) – 9 items	62.57 (23)	83.52 (17)***	.97	.95	.07	.04	106.57	190.15
Model 5. Four factors (BA, FA, COH and SCO) – 11 items	111.27(38)	48.70 (14)***	.95	.93	.08	.04	169.27	279.44

*** p<.001

The results of the multi-group analysis, to test the invariance of the factor structure, showed that the model is strongly invariant across countries (ΔCFI=.008; ΔRMSEA=.001), whereas not only strong invariance, but also strict invariance (factor loadings, item intercepts, and item uniqueness invariance) was found across genders (ΔCFI=.004; ΔRMSEA=.006). A failure of the calculations to converge meant that we were unable to test the invariance across the stimulus images

In order to assess whether the PRS-11 discriminates among environmental categories, a one-way ANOVA were performed. This analysis showed that the PRS-11 scores of the five environmental categories differ from one another [$F_{(4,330)}$=24.474; p<.001; effect size was moderate: η^2=.23]. Post-hoc comparisons (with Bonferroni correction) showed that hills and lakes had the highest PRS scores, followed by housing and city streets with significantly lower PRS scores. Industrial zone had the lowest PRS score.

4. Conclusion and discussion

In case of self-rating instruments one could argue that when research subjects use the scale to rate the same stimulus, they respond in various ways depending on their prior experiences, their interpretation of the wording, and the stimulus attributes. In contrast, ART assumes our "prior experience" in relation to natural stimuli to be the same (see Wilson's biophilia hypothesis, 1984; Kaplan's psycho-evolutionary perspective, in Kaplan & Kaplan, 1989): From the Wilson and Kaplan perspective, human beings are biologically organised to respond to natural environments in a direct way, i.e. by experiencing and reporting strong preferences and high levels of potential restorativeness. While developing a short form of the PRS, we have shown that the semantic aspects of a scale item make a substantial difference in the way people assess environmental stimuli. Importantly, we have demonstrated that beginning with cognitive interviewing do develop an in-depth understanding of the meaning of each item resulted in a scale that was independent of subject's gender and nationality. More research is needed in this direction to define an instrument completely independent from stimulus attributes as well. This kind of independence is fundamental for a self-rating scale to be a good measurement instrument.

Appendix A.

The PRS-11 (in brackets the original number and factor in Hartig et al.'s 1997a scale)

Fascination

Places like that are fascinating (FA 12)

In places like this my attention is drawn to many interesting things (FA 7)

In places like this it is hard to be bored (FA 11)
Being Away
Places like that are a refuge from nuisances (B-A 1)
To get away from things that usually demand my attention I like to go to places like this (B-A 5)
To stop thinking about the things that I must get done I like to go to places like this (B-A 4)
Coherence
There is a clear order in the physical arrangement of places like this (COH 15 Rev)
In places like this it is easy to see how things are organised (COH 26)
In places like this everything seems to have its proper place (new item)
Scope
That place is large enough to allow exploration in many directions (FA 10)
In places like that there are few boundaries to limit my possibility for moving about (new item)

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The development of a six-item short-form of the state scale of the Spielberger State–Trait Anxiety Inventory (STAI)

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Two studies are reported describing the development of a short-form of the state scale of the Spielberger State–Trait Anxiety Inventory (STAI) for use in circumstances where the full-form is inappropriate. Using item–remainder correlations, the most highly correlated anxiety-present and anxiety-absent items were combined, and correlated with scores obtained using the full-form of the STAI. Correlation coefficients greater than .90 were obtained using four and six items from the STAI. Acceptable reliability and validity were obtained using six items. The use of this six-item short-form produced scores similar to those obtained using the full-form. This was so for several groups of subjects manifesting a range of anxiety levels. This short-form of the STAI is therefore sensitive to fluctuations in state anxiety. When compared with the full-form of the STAI, the six-item version offers a briefer and just as acceptable scale for subjects while maintaining results that are comparable to those obtained using the full-form of the STAI.

The Spielberger State–Trait Anxiety Inventory (STAI) is one of the most frequently used measures of anxiety in applied psychology research. It is a reliable and sensitive measure of anxiety. Its popularity has meant that researchers are able to compare their results with those of others, a useful but rare occurrence in such research. The one drawback, however, with the STAI is its length, being 40 items long. This is a barrier to its use in studies where time for completing questionnaires is limited. It is also a barrier for some populations who are unable or unprepared to complete lengthy scales with seemingly repetitive items. These problems could be largely overcome by the development of a standardized short-form of the STAI.

The STAI consists of two questionnaires of 20 items each. The first questionnaire measures state anxiety (how one feels at the moment), the second, trait anxiety (how one generally feels). There are two STAI questionnaires: one produced in 1970 (Form X), the other in 1983 (Form Y). As yet there is no standardized short-form of the STAI. Some researchers (Foa, McNally & Murdock, 1989; McNally, Foa & Donnell, 1989; Smith, Ingram & Brehm, 1983) have used a 10-item STAI. The 10 items selected are alternate anxiety-present and anxiety-absent items from the full-form of the STAI. Where time is a consideration, Spielberger recommends using the 10-item anxiety score of the State–Trait Personality Inventory (STPI, 1979). None of these shortened forms of the STAI has been standardized.

* Requests for reprints.

The aim of the two studies reported here is to develop a short-form of the state measure of the STAI for use in settings where time constraints preclude the use of the full-form.

STUDY 1: ITEM SELECTION

The purpose of this study is to select the smallest subset of both anxiety-present and anxiety-absent items from the full-form of the STAI which is highly correlated (i.e. $r > .90$) with scores obtained using the full-form of the STAI. Spielberger (1983) states that equal numbers of anxiety-present and anxiety-absent items constitute a more stable and consistent measure.

Method

Subjects

The sample comprised 200 pregnant women, from whom completed STAIs had been obtained as part of another study (Marteau, Johnston, Kidd, Michie, Cook, Slack & Shaw, 1992).

Procedure

The questionnaires were completed while women were waiting for a routine out-patient appointment. Scores on each item were correlated with scores on the remaining 19 items of the STAI (item-remainder correlations), to assess the contribution of each item to the overall state anxiety score.

Results

The items were ranked in order of magnitude of their item-remainder correlations (see Table 1). Equal numbers of anxiety-present and anxiety-absent items were then used to form two-, four-, six-, eight- and ten-item forms of the STAI. The correlations between scores from these short-forms and the full length STAI were: $r = .96$ for the 10 items; $r = .96$ for the eight items; $r = .95$ for the six items; $r = .91$ for the four items; and $r = .84$ for the two items.

Table 1. Item–remainder correlations: Pearson correlation coefficients between each item and the total of the remaining items of the STAI

Item no.	Adjective	*r*	Item no.	Adjective	*r*
1	*calm*	*.71*	11	self-confident	.50
2	secure	.66	12	nervous	.42
3	*tense*	*.62*	13	jittery	.48
4	regretful	.37	14	high-strung	.50
5	at ease	.66	15	*relaxed*	*.71*
6	*upset*	*.53*	16	*content*	*.69*
7	misfortunes	.51	17	*worried*	*.58*
8	rested	.42	18	over-excited	.33
9	anxious	.49	19	joyful	.40
10	comfortable	.58	20	pleasant	.63

Note. Italics depict the three highest anxiety-present and anxiety-absent items.

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Discussion

Correlations between subsets of four and six items of the STAI were greater than $r = .90$. Whether either is an acceptable short-form of the STAI will depend upon the extent to which they are reliable and valid in comparison with the full-form of the STAI.

STUDY 2: RELIABILITY AND VALIDITY

The aim of this study is to determine the reliability and validity of the four- and six-item short-forms of the STAI as a basis for choosing the optimal short-form. Internal reliability was assessed. Test–retest reliability was not calculated for, as Spielberger notes, test–retest reliability is low for state anxiety because of state anxiety's transitory nature (Spielberger, 1983). The form of validity for a short-form of the STAI to be assessed is concurrent validity, i.e. the extent to which scores on the short-form are similar to those obtained using the full-form. This is assessed for groups with mean STAI ratings near the population mean (unselected pregnant women), as well as groups with known raised anxiety (pregnant women following receipt of an abnormal test result during pregnancy).

Method

Subjects

The subjects comprised three groups: Group 1: 38 medical students, attending a psychology lecture; Group 2: 45 student nurses, attending a nursing lecture; Group 3: 200 pregnant women, as in Study 1, above; and Group 4: 23 pregnant women who had received an abnormal result on a routine screening test for fetal abnormality. Data from these women had been obtained as part of another study (Marteau, Cook, Kidd, Michie, Johnston, Slack & Shaw, in press).

Procedure

Questionnaires were completed while subjects waited for lectures (Groups 1 and 2), awaited a routine clinic appointment (Group 3), and at home, using a postal questionnaire (Group 4).

Results

The reliability coefficient for the 20-item STAI was: $\alpha = .91$; for the six items: $\alpha = .82$; for the four items: $\alpha = .77$.

Validity

The comparability of prorated scores from the six- and four-item scales with the use of the full scale was used to determine the concurrent validity of the two short-forms of the STAI. To avoid over-inflated correlations caused by items correlating with themselves (Pierce, Sarason & Sarason, 1986), the six-item questionnaire was compared with both the 20- and 14-item questionnaires, and the four-item questionnaire was compared with both the 16- and 20-item questionnaires. Comparisons were made using paired t tests (see Table 2).

There were no differences in the mean scores obtained using the full-form of the STAI and those obtained using the prorated six-item short-form, for any of the three

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Table 2. STAI scores (mean SD) derived from (*a*) the full-form, the prorated six-item short-form and the prorated 14-item short-form and (*b*) the full-form, the prorated four-item short-form and the prorated 16-item short-form

	Pregnant women ($N = 200$)	Student nurses ($N = 45$)	Medical students ($N = 38$)
(a)			
Full-form (20 items)	37.6 (9.1)	39.7 (7.0)	46.2 (6.2)
Prorated (6 items)	37.1 (11.0)	39.9 (7.5)	46.8 (6.2)
Prorated (14 items)	37.5 (8.7)	39.8 (6.9)	46.2 (7.1)
(b)			
Full-form (20 items)	37.6 (9.1)	39.7 (7.0)	46.2 (6.2)
Prorated (4 items)	38.4 (12.0)	41.4 (8.8)	48.6 (7.1)
Prorated (16 items)	36.9 (8.7)**	39.4 (7.0)*	45.6 (6.7)*

Note. Paired *t* tests were undertaken between data obtained from the prorated 6- (4-) item forms and both the full-form and prorated 14- (16-) item form.
* $p < .05$; ** $p < .01$.

groups. Similarly, there were no differences between the means prorated from the six-item short-form and the means prorated from the remaining 14 items. The prorated means obtained from the four item short-form were all significantly higher than the means based on the 16 remaining items and the full 20 items. Given this evidence suggesting the validity of the six-item short-form, further analyses were conducted with just this short-form of the STAI.

To determine whether the six-item short-form of the STAI is sufficiently sensitive to detect the changes in anxiety detected by the full-form, STAI scores obtained using the six-item short-form were compared with those obtained using the full-form on a sample with known elevated anxiety. Following receipt of an abnormal result on maternal serum alpha-fetoprotein screening indicating increased risk of spina bifida and Down's syndrome, women's anxiety scores are significantly raised compared both with their levels prior to testing, and when compared with women receiving a normal test result (Marteau *et al.*, in press). The mean STAI score for these women immediately following receipt of an abnormal test result is 46.4 (SD 14.8); the mean score prorated from the six-item short-form is 47.7 (SD 15.8): $t(22) = 0.96$, $p = .35$). Thus, the mean score obtained from the six-item short-form is similar to that obtained using the full-form of the STAI.

General discussion

The six-item short-form of the STAI (STAI-6: Appendix A) produces scores similar to those obtained using the full 20-item STAI. It has acceptable reliability, and produces scores that are similar to those produced with the full-form across subject groups manifesting normal and raised levels of anxiety. Two of the three anxiety-absent items are those identified by Spielberger to be particularly sensitive to low

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stressors; all three anxiety-present items he reports as being particularly sensitive to high stressors (Spielberger, 1983). When compared with the full-form of the STAI, the six-item short-form offers a briefer and more acceptable scale for subjects; it maintains results that are comparable to those obtained using the full-form of the STAI and remains sensitive to different degrees of anxiety.

There are several advantages to using a brief questionnaire. It is likely to maximize response rates, and minimize the number of response errors and unanswered items, thus improving the validity and generalizability of any findings. The longer a questionnaire, the more the responding sample will be biased in favour of those who find reading and writing easy. A brief questionnaire will also reduce the time taken to complete the questionnaire, and the time taken to score the results.

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Appendix A: Example from the Self-Evaluation Questionnaire (STAI-AD)

[Correction added on 20 December 2019 after first online publication: Appendix questionnaire has been updated.]

Directions

A number of statements which people have used to describe themselves are given on the following pages. Read each statement and then select the appropriate button to indicate how you feel ***right*** now, that is, ***at this moment***. There are no right or wrong answers. Do not spend too much time on any one statement but give the answer which seems to describe your present feelings best.

Not at all - Somewhat - Moderately so - Very much so



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Outdoor Thermal Comfort at a University Campus: Studies from Personal and Long-Term Thermal History Perspectives

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Abstract: Thermally comfortable outdoor spaces have contributed to high-quality urban living. In order to provide a further understanding of the influences of gender and long-term thermal history on outdoor thermal comfort, this study conducted field surveys at a university campus in Shanghai, China by carrying out microclimatic monitoring and subjective questionnaires from May to October, 2019. The analysis of collected data found that, during our survey, 57% of the occupants felt comfortable overall and 40–60% of them perceived the microclimate variables (air temperature, humidity, solar radiation, and wind speed) as "neutral". The universal thermal climate index (UTCI) provided a better correlation with occupant thermal sensation than the physiologically equivalent temperature (PET). Females were more sensitive to the outdoor thermal environment than males. Older age led to lower thermal sensation, but the thermal sensitivities for age groups of <20, 20–50, and >50 were similar. Occupants who had resided in Shanghai for a longer period showed higher overall comfort rating and lower thermal sensation. Interviewees who came from hot summer and cold winter climate regions were less effected by the change of UTCI than those from severe cold or cold climate regions.

Keywords: outdoor thermal comfort; field survey; urban open spaces; microclimate; personal factors; long-term thermal history; Universal Thermal Climate Index (UTCI)

1. Introduction

With 56% of the global population living in cities in 2020 [1], thermal comfort in urban outdoor spaces has gained growing research attention worldwide. Thermally comfortable urban open spaces offer high-quality locations for various activities, attract citizens to the outdoors, and improve the vitality of the city [2–8]. Increased outdoor activity may also help in saving building energy consumption [9–12]. According to the review by Lai et al. [13], field studies concerning outdoor thermal comfort have been conducted in more than 100 cities globally to gain a better understanding of this topic. These studies have demonstrated that physical, physiological, and psychological factors are direct influencing factors of outdoor thermal comfort.

From a physical perspective, many studies [14–18] have demonstrated that the physical environment parameters such as air temperature, wind, thermal radiation, and humidity are crucial because they determine the heat transfer between a human body and the outdoor thermal environment [19]. Besides physical parameters, physiological signals of the human body such as skin temperature and rectal temperature were shown to be a good indicator of outdoor thermal comfort [20–22]. Psychological influences [23] include experience, expectation, and perceived control, and these altered people's thermal sensation in outdoor spaces.

Although the direct influencing factors (physical, physiological, and psychological) are known, outdoor thermal comfort remains a complex issue. For example, due to individual differences, subjects of different genders and ages may have distinctive thermal perceptions, even under the same thermal environment. Females were often found to be more sensitive to the thermal environment than males [24–26], while the elderly group was the least sensitive compared to other age groups [27]. However, the findings were not consistent. Cohen et al. [28] demonstrated that females had a higher tolerance to cold than males did. In addition to personal factors, people's long-term thermal history affected their current thermal sensation. Some studies [29–31] have reported that local residents expressed higher satisfaction to the outdoor thermal environment than visitors did, but the researchers did not investigate the effect of former climatic experiences.

The contradictory findings regarding personal influences on outdoor thermal comfort and the insufficient research effort on the effect of long-term thermal history suggest that further investigation should be conducted. This paper reports the findings of a thermal comfort field survey of a university campus. One main focus of this study is to analyze the influences from a personal and long-term thermal history perspective.

2. Methods

A field survey was conducted at a university campus to study the outdoor thermal comfort. This section first describes the investigated site. Then, the field survey is presented in detail. Finally, the thermal indices used for the data analysis are introduced.

2.1. Site Description

The investigation was conducted at the Minhang Campus of Shanghai Jiao Tong University in Shanghai, China. We chose to perform the outdoor thermal comfort field study at a university campus because the majority of interviewees were university students, and they came from different climate zones and had distinctive thermal histories. The campus has an area of 309 hectares and encounters a wide variety of thermal environments. As shown in Figure 1, the field survey was performed at three sites with a distinctive surrounding morphology. The first site is a grand lawn with an open view and is exposed to direct sun and wind. The second site is a lawn near a lake, but it is much smaller compared to the first site. The third site is close to the second one, but it has clusters of trees, which provide shade. The users of the investigated sites performed various activities such as exercising, relaxing, reading, and chatting.

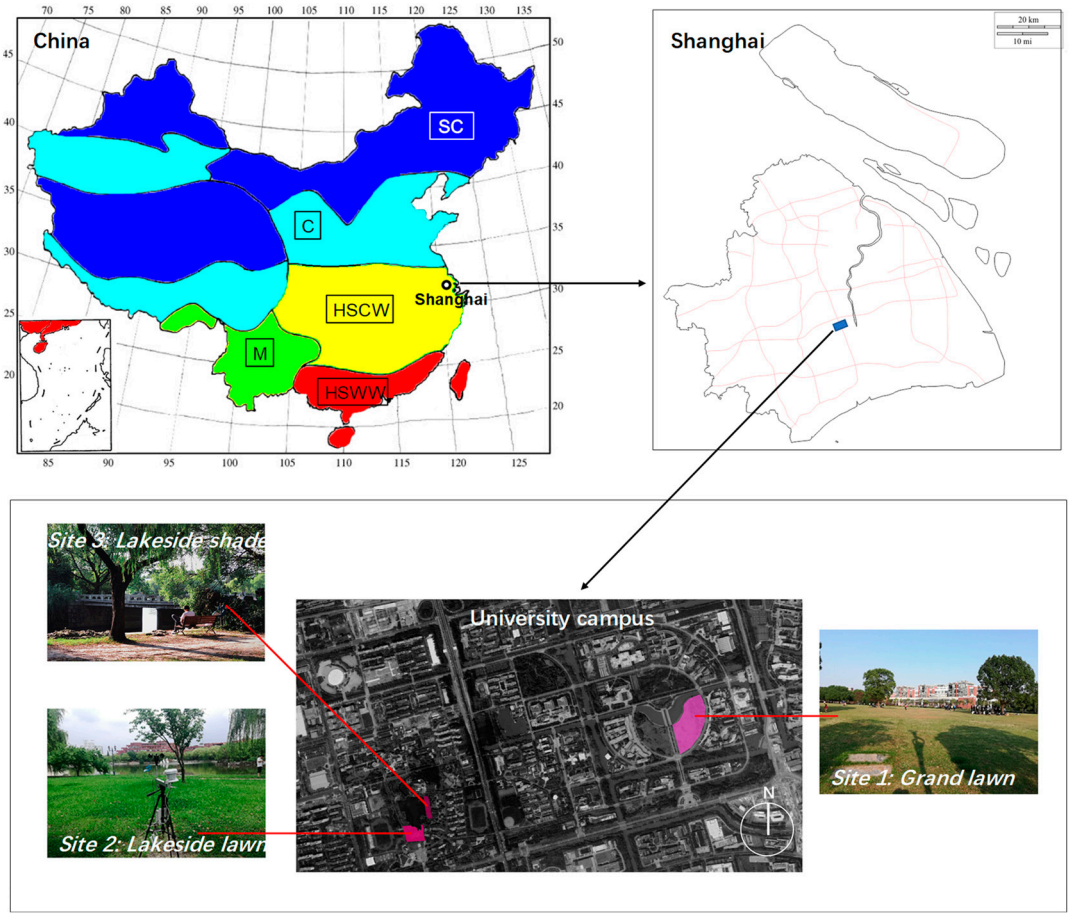


Figure 1. China (with climate zones), Shanghai, and the investigated sites at the university campus.

2.2. Field Survey

In order to collect data to analyze outdoor thermal comfort, this investigation conducted six field surveys from 1 May 2019 to 3 November 2019. During the field survey, subject interviews were conducted while, simultaneously, the microclimate was monitored. For the microclimate monitoring, portable weather stations were deployed at the sites to record air temperature (T_a), relative humidity (RH), wind speed (V_a), and global solar radiation (G) at a height of 1.5 m above the ground. Table 1 shows the specifications of the sensors used to record the microclimatic variables.

Table 1. Sensors used in this study and their specifications.

Parameter	Sensor	Range	Accuracy
Air temperature	S-THB-M002	−40–75 °C	±0.2 K at 20 °C
Relative humidity	S-THB-M002	0–100%	±3%
Wind speed	S-WSET-A	0–45 m/s	±1.1 m/s
Global radiation	S-LIB-MOO3	0–1280 W/m^2	±10 W/m^2 or ±5%

A voluntary questionnaire was distributed to occupants in the studied sites to collect subjective data. The collected information included personal data such as gender, age, clothing insulation, activity level, and long-term thermal history information such as length of residence in Shanghai and former residence. Subjective perception and preference of the thermal environment were also recorded. The perceptions included seven-point thermal sensation [32], and those of the wind, sun, and humidity. The preference questions asked the occupants if they would like the microclimatic parameters (air temperature, wind, solar radiation, humidity) to be higher, unchanged, or lower. The overall comfort was rated on a five-point scale: comfortable, slightly uncomfortable, uncomfortable,

very uncomfortable, and intolerable. The questionnaire used in the investigation is shown in Figure 2. A total of 520 samples were obtained in the field survey, with 75 samples in May, 136 in June, 118 in July, and 191 in October.

Outdoor Thermal Comfort Questionnaire

Date: ____/____/____ Time: ____:____ Location:_____

Gender: Male/Female Age:_____

Your hometown (City, Country):

How long have you lived in Shanghai?
A. Less than 1 year B.1-3 years C. 3-7 years D. More than 7 years

Your current activity:
A. Exercising (Light), B. Exercising (Medium), C. Exercising (Heavy), D. Chatting (Standing), E. Chatting (Seated), F. Strolling, G. Resting, H. Attending Children

What are you wearing right now?
A. T-Shirt (Short sleeves), B. T-Shirt (Long sleeves), C. Shorts OR short skirt,
D. Long Pants OR long skirt, E. Vest, F. Sport Skirt, G. Jacket
If you are wearing other clothing, please indicate: _____

Please describe your current thermal sensation:

Cold	Cool	Slightly Cool	Neutral	Slightly Warm	Warm	Hot

How do you perceive the following meteorological parameters at this moment?

Wind Speed	Very weak	Weak	Slightly weak	Neutral	Slightly strong	Strong	Very strong
Humidity	Very dry	Weak	Slightly dry	Neutral	Slightly damp	Damp	Very damp
Solar Radiation	Very weak	Weak	Slightly weak	Neutral	Slightly strong	Strong	Very strong

What are your preferences in regard to the following meteorological parameters:

Temperature	Higher	Unchanged	Lower
Wind Speed	Stronger	Unchanged	Weaker
Humidity	Damper	Unchanged	Drier
Solar Radiation	Stronger	Unchanged	Weaker

Please describe your overall comfort level:

Intolerable	Very uncomfortable	Uncomfortable	Slightly uncomfortable	Comfortable

Figure 2. Thermal comfort questionnaire used in this study.

2.3. Thermal Indices

Two thermal indices were chosen for the analysis of outdoor thermal comfort: Physiologically Equivalent Temperature (PET) [33] and Universal Thermal Climate Index (UTCI) [34]. The two indices assess the influences from the outdoor thermal environment parameters in an integrated manner, and they have been widely used in outdoor thermal comfort studies. In this study, PET and UTCI were calculated using the RayMan program [35,36].

2.3.1. Physiologically Equivalent Temperature (PET)

The PET index is an equivalent temperature based on the two-node Munich Energy-balance Model for Individuals [37]. PET is defined as the air temperature of a typical indoor room producing the same core and skin temperature as the actual complex outdoor conditions.

2.3.2. Universal Thermal Climate Index (UTCI)

UTCI is similar to PET, but the UTCI has several advantages. For example, UTCI is based on an advanced multi-node human thermoregulation model [38]. In addition, UTCI considers the changes in clothing insulation with the changes in ambient temperature [38]. When comparing the typical indoor and actual environments, UTCI uses integrated physiological signals as criteria [39].

3. Results

Because the microclimatic parameters are essential factors in outdoor thermal comfort [40], this section first presents the weather of Shanghai and the microclimatic conditions present during the survey. Then, thermal comfort is demonstrated descriptively and analyzed with respect to PET and UTCI. Finally, the influences of personal factors and long-term thermal history were studied and presented.

3.1. Climate and Microclimate

According to Koppen Climate classification [41], Shanghai has a humid subtropical climate (Csa) with a hot summer and a cold winter, and it is generally moist throughout the year. Figure 3 shows the mean, averaged maximum, and averaged minimum monthly air temperature and mean monthly relative humidity in Shanghai according to Chinese Standard Weather Data (CSWD) from 1971 to 2003 [42]. The hottest month was July, with a monthly mean T_a of 27 °C, while the coldest month was January, with a 4 °C monthly mean T_a. The relative humidity was constantly high, exceeding 65% for all months. In summer months (June, July, and August), the mean monthly RH was greater than 80%. The wind speed 10 m above ground was generally higher than 3 m/s.

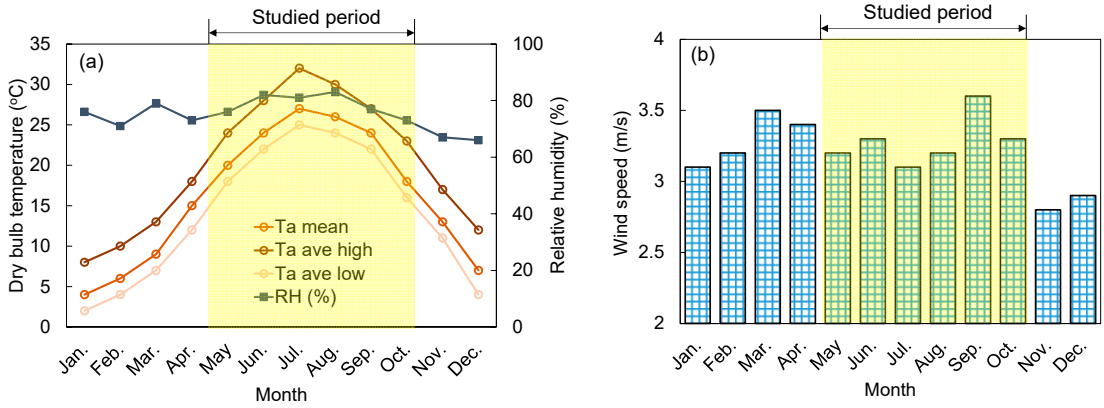


Figure 3. General climate of Shanghai recorded at a rural station 20 km from the campus: (**a**) mean, averaged maximum, averaged minimum air temperature, mean monthly relative humidity, and (**b**) mean wind speed at 10 m above ground [42].

Our field investigation was conducted from May to October, when the weather ranged from mild to hot. Table 2 shows the microclimatic conditions measured at the three sites during the survey. The air temperature was between 20.7 and 33.1 °C, while the relative humidity ranged from 46.6% to 79.7%. The different forms of the three sites have led to distinct wind speed and solar radiation levels. With its wide-open form, the grand lawn had the highest mean wind speed of 0.9 m/s and mean global radiation of 239 W/m^2, followed by the lakeside lawn (0.4 m/s and 232.4 W/m^2). The shaded place had compact surroundings, so the mean wind speed was only 0.2 m/s and the mean global radiation was 100.2 W/m^2.

Table 2. Microclimate thermal conditions during the survey.

		Air Temperature (T_a, °C)	Relative Humidity (RH, %)	Wind Speed (V_a, m/s)	Global Radiation (G, W/m^2)
Site 1: Grand lawn (May, June, July, October)	Mean	24.5	68.3	0.9	239.0
	Max.	33.1	79.7	3.5	1115.6
	Min.	19.2	46.6	0.0	1.9
Site 2: Lakeside lawn (June, July, October)	Mean	27.3	64.9	0.4	232.4
	Max.	32.8	78.4	2.0	1025.6
	Min.	20.9	47.3	0.0	13.1
Site 3: Lakeside shade (June, July, October)	Mean	24.7	63.0	0.2	100.2
	Max.	31.8	78.8	1.5	745.6
	Min.	20.7	46.8	0.0	3.1

3.2. Demographics of the Interviewees

Because the major objective of this study is to understand the influence of personal factors and long-term thermal history on outdoor thermal comfort, we summarized the gender, age, length of residence, and hometown climate of the respondents in Figure 4. According to the survey results, our interviewees consisted of more males (57%) than females (43%). Figure 4b shows that the age of the respondents was concentrated in the range of 10 to 40. The residence length was more evenly distributed than age; Figure 4c shows that 22%, 30%, 16%, and 32% of the interviewees have lived in Shanghai for less than 1 year, 1 to 3 years, 3 to 7 years, and more than 7 years, respectively. The climate of the hometown of the respondents was classified into severe cold (SC), cold (C), hot summer and cold winter (HSCW), hot summer and warm winter (HSWW), and mild (M) based on average air temperatures in the coldest and hottest months of the year [43], as shown in Figure 1. It is noteworthy that Shanghai was categorized as a HSCW climate. The distribution of hometown climate was uneven, with 69% of the respondents coming from a HSCW region. C regions accounted for 19% of respondents, but the other three climate zones represented a share of less than 5%.

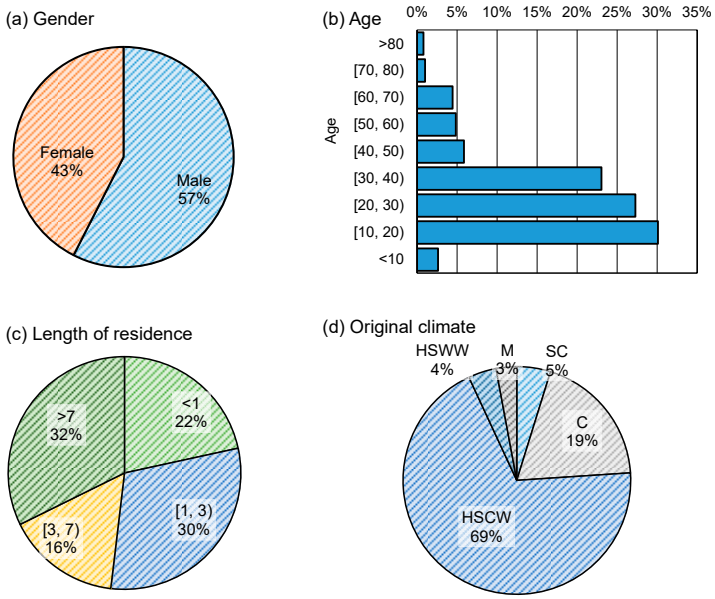


Figure 4. An overview of the demographics of respondents: (**a**) gender, (**b**) age, (**c**) length of residence, and (**d**) hometown climate: severe cold (SC), cold (C), hot summer and cold winter (HSCW), hot summer and warm winter (HSWW), and mild (M).

3.3. General Description of Thermal Comfort

In Figure 5, we first provide an overall picture of our survey results by demonstrating the distributions of the perception vote, including the thermal sensation vote (TSV), the humidity sensation vote (HSV), the sun sensation vote (SSV), and the wind sensation vote (WSV). The "neutral" perception accounted for the highest proportion among all scales for all four microclimatic variables. The distributions of the perception votes followed a normal distribution except for the TSV. For the TSV, the "hot" vote accounted for 15% of responses. The humidity sensation vote was slightly skewed to the "damp" side, which reflected the natural humidity of Shanghai's climate.

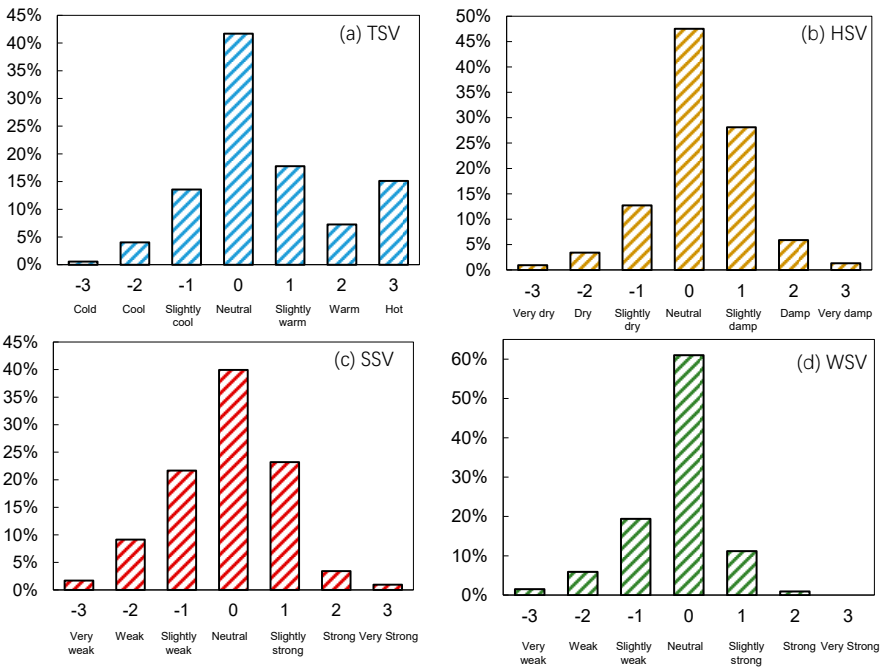


Figure 5. Distributions of: (**a**) thermal sensation vote (TSV), (**b**) humidity sensation vote (HSV), (**c**) sun sensation vote (SSV), and (**d**) wind sensation vote (WSV).

Figure 6 shows the distributions of the preferences for air temperature, humidity, solar radiation, and wind speed. Most participants voted for the "unchange" option for the four microclimatic parameters, showing a generally high acceptance of the outdoor thermal environment during our survey. Figure 6 also demonstrates that the general climate was warm because the percentages of respondents who wanted the air temperature, humidity, and solar radiation to be "lower" were larger than the percentages of the "higher" vote, and the wind speed had a reversed trend. The result of preference vote distributions agreed with the climate measured during the survey (Table 2).

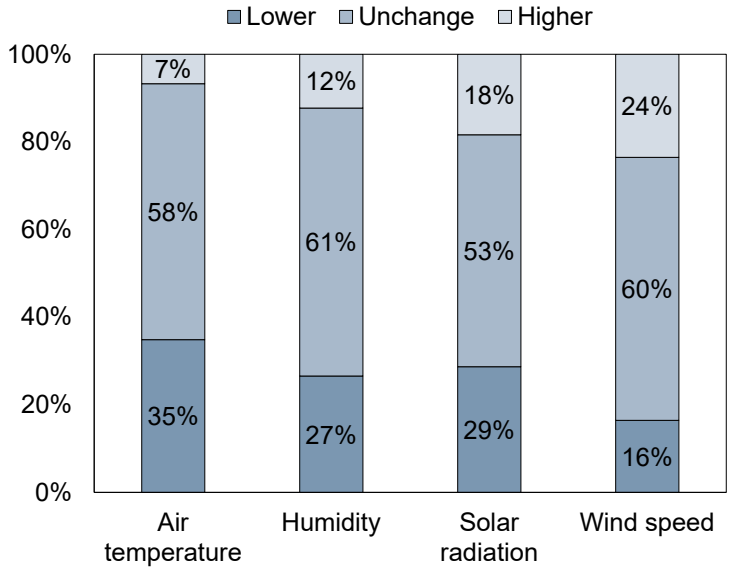


Figure 6. Distributions of preference votes for air temperature, humidity, solar radiation, and wind speed.

Figure 7 demonstrates the distribution of overall comfort. Overall, 57% of the respondents indicated that they felt comfortable. Only 1%, 3%, and 8% of them regarded the outdoor thermal environment as intolerable, very uncomfortable, or uncomfortable, respectively.

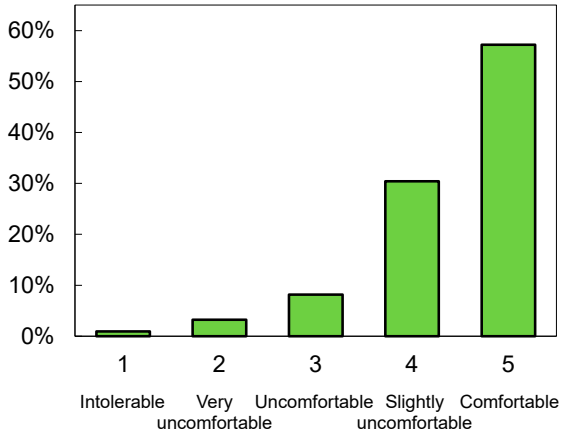


Figure 7. Thermal comfort vote distribution.

Figure 8 examines the overall comfort rating at different thermal sensation levels. Due to the insufficient "cold" vote (N = 3), this level was excluded from the analysis. We have assigned numerical values 1 to 5 to intolerable, very uncomfortable, uncomfortable, slightly uncomfortable, and comfortable, respectively. A higher score indicates higher overall comfort perception. A decrease in overall comfort was observed from "neutral" to "hot". However, it is interesting to find that "cool", "slightly cool" and "neutral" thermal sensations had similar overall comfort ratings. This is because our survey was

conducted in mild-to-hot seasons, and previous studies [44,45] have shown that, in hot seasons, people perceived "cool" sensations as more comfortable than "warm" ones.

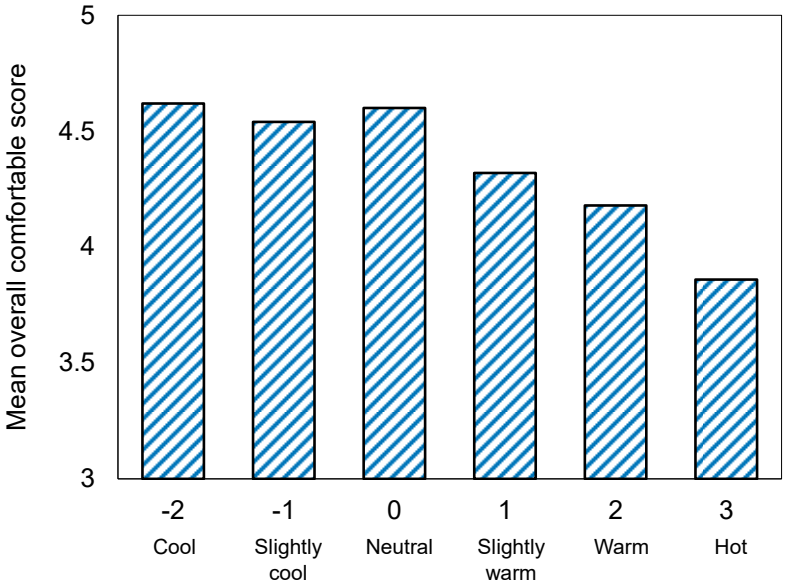


Figure 8. Mean overall comfort score for different levels of thermal sensations.

3.4. Correlations between Perceptions and Thermal Indices (PET and UTCI)

Our study has calculated the PET and UTCI values for each observation obtained during the field survey and used the "bin method" [46] to calculate the mean thermal sensation for every 1 K PET or UTCI interval, and the results are plotted in Figure 9 with linear regression applied.

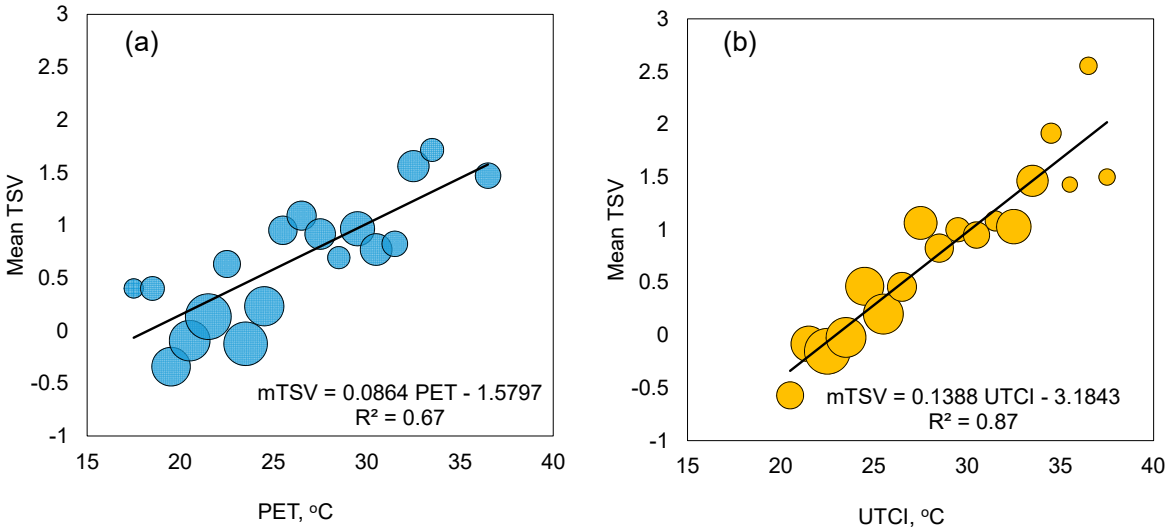


Figure 9. Relationships between: (**a**) physiologically equivalent temperature (PET) and mean TSV, (**b**) Universal Thermal Climate Index (UTCI) and mean TSV, for all genders and all age groups.

From the R^2 values, it can be seen that UTCI provided a better correlation with the mean TSV than PET did. It is not surprising to see that thermal sensation became greater at higher thermal indices. In addition, the slopes for the PET and UTCI regression equations were 0.0864 and 0.1388, which correspond to 11.6 PET/TSV and 7.2 UTCI/TSV, respectively.

Figure 10 depicts the relationships between overall comfort and the thermal indices using a similar method to that applied in Figure 9. People perceived a worse overall comfort as the thermal indices increased. By comparing the values of R^2 between Figures 9 and 10, poorer association was found

for overall comfort than thermal sensation. The UTCI also showed a better correlation with overall comfort than PET. Due to the similar trend between UTCI and PET and the higher R^2 values of UTCI compared to PET, the subsequent analysis used UTCI to further investigate the influences of personal factors and long-term thermal history.

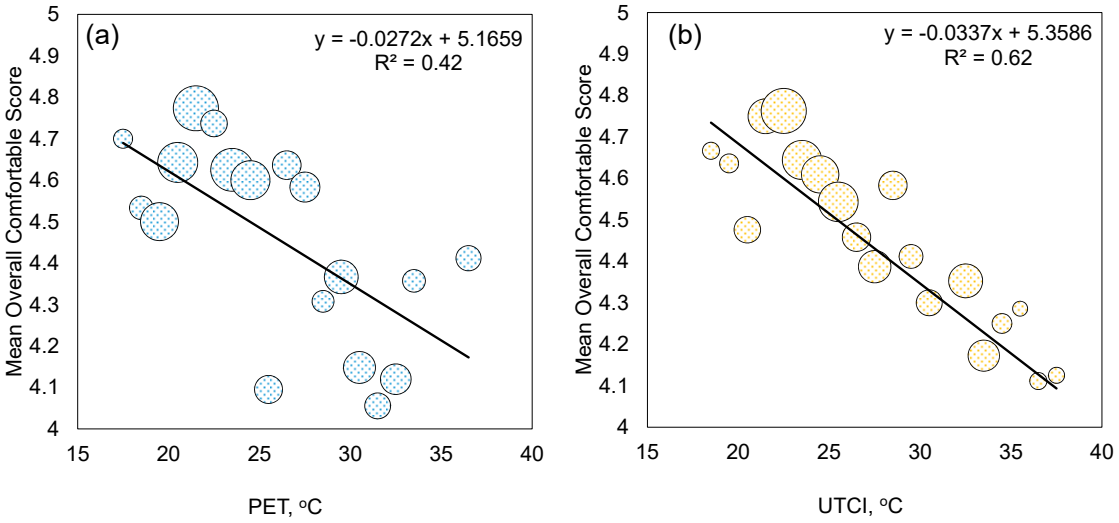


Figure 10. Relationships between: (**a**) PET and mean overall comfort, (**b**) UTCI and mean overall comfort, for all genders and all age groups.

3.5. Influence of Personal Factors

The personal influence was investigated by analyzing the effect of gender and age. Figure 11 uses the "bin method" to plot the mean TSV for every 2-K interval of UTCI for male and female groups. Splitting the data according to gender reduced the number of observations in each data pool. Thus, the 2-K interval was used instead of 1 K to increase the sample size within each interval.

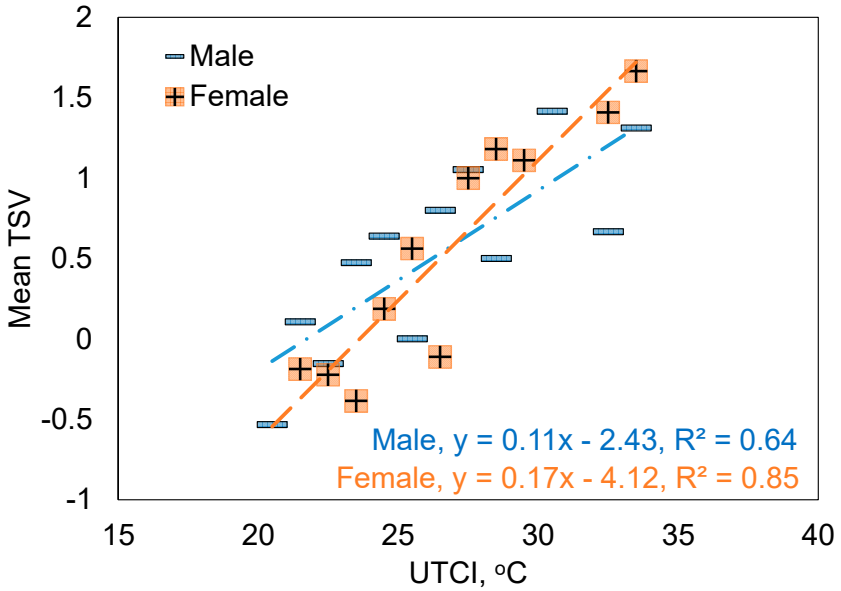


Figure 11. Relationships between mean TSV and UTCI for males and females.

As shown in Figure 11, the slopes for the linearly regressed equations for males and females were 0.11 and 0.17, respectively, showing that males were less thermally sensitive to the outdoor thermal environment than females were the two equations intersect at around 27 °C UTCI. This shows that females in our study felt warmer than males when the UTCI was above 27 °C UTCI, and had a lower TSV than males when UTCI was lower than 27 °C UTCI.

To study the effect of age on outdoor thermal comfort, the dataset was divided into three age groups: younger than 20, from 20 to 50, and older than 50. The "Bin method" was applied to the three age groups, and the mean TSV was calculated for every 2-K interval of UTCI, as shown in Figure 12. Bins with a sample size of less than five were discarded from the analysis. A distinct trend was demonstrated among age groups. While the thermal sensitivity for different groups was similar, notable TSV differences can be found in Figure 12. The <20-years-old age group had the highest thermal sensation while the occupants aged >50 years demonstrated the lowest TSV. It is worth noting that, due to the limited sample size of the elderly age group, the result of the >50 group was concentrated at a lower air temperature region (<26 °C UTCI), and the result may not be representative.

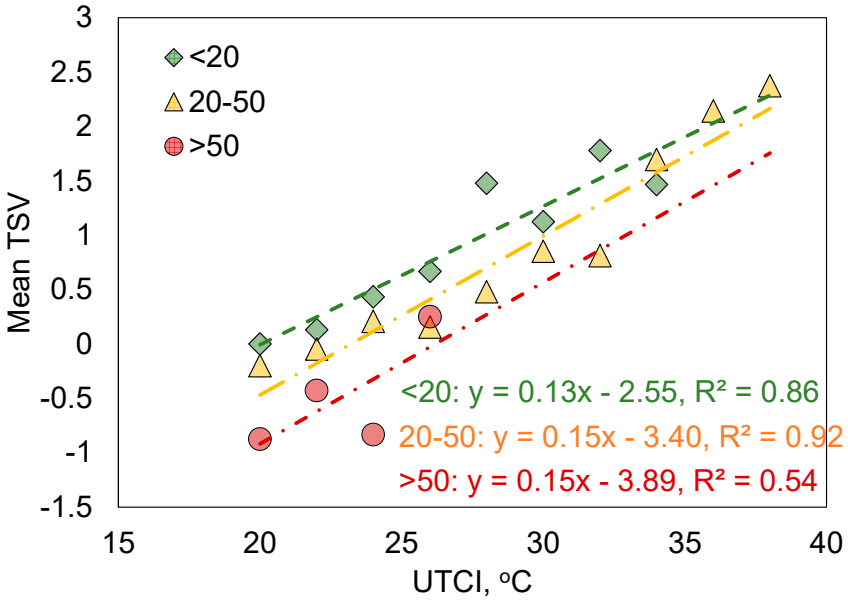


Figure 12. Relationships between mean TSV and UTCI for ages of <20, 20–50, and >50 years.

3.6. Influence of Long-Term Thermal History

The influence of long-term thermal history was analyzed by examining the effect of local residence length and hometown climate. Figure 13 demonstrates the distributions of the overall thermal comfort vote for occupants with different lengths of residence in Shanghai. With longer residence duration, occupants became better adapted to the local thermal environment and had a higher overall comfort level. Figure 14 further plots the relationships between UTCI and mean TSV for respondents with different residence lengths. The general pattern shows that longer residence led to a lower TSV during the studied period of our survey.

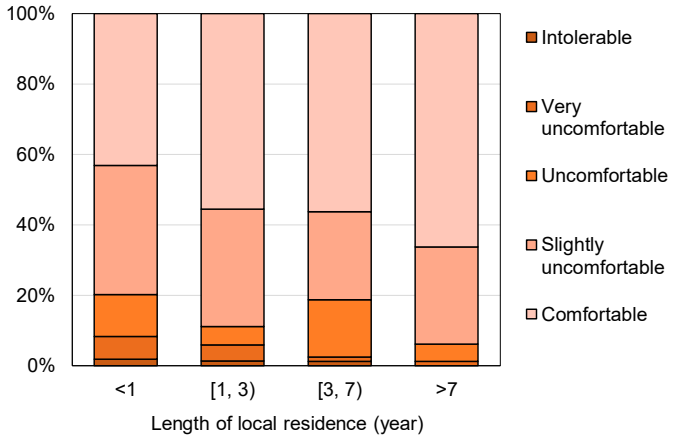


Figure 13. Distributions of overall comfort vote for respondents with different residence lengths in Shanghai.

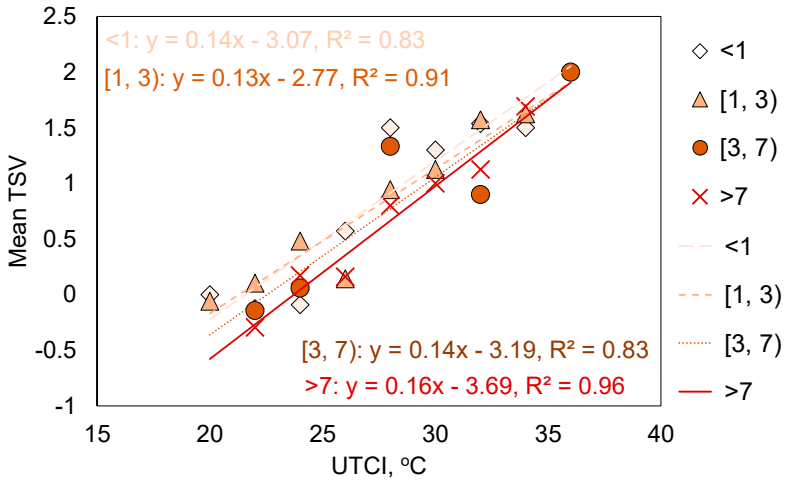


Figure 14. Relationships between UTCI and mean TSV for interviewees with different lengths of residence.

The observations collected from severe cold (SC), hot summer and warm winter (HSWW), and mild (M) climate zones accounted for less than 5% of the total samples. The analysis was not performed on HSWW and M zones due to the insufficient size of the sample. The data from the SC region were combined with those of the C region to further compare with the data from the hot summer and cold winter (HSCW) region. The HSCW and SC/C regions had 360 and 124 samples, respectively. Figure 15 demonstrates the mean TSV as a function of UTCI for SC/C and HSCW. Respondents from HSCW were less sensitive to UTCI than respondents from the SC and C regions.

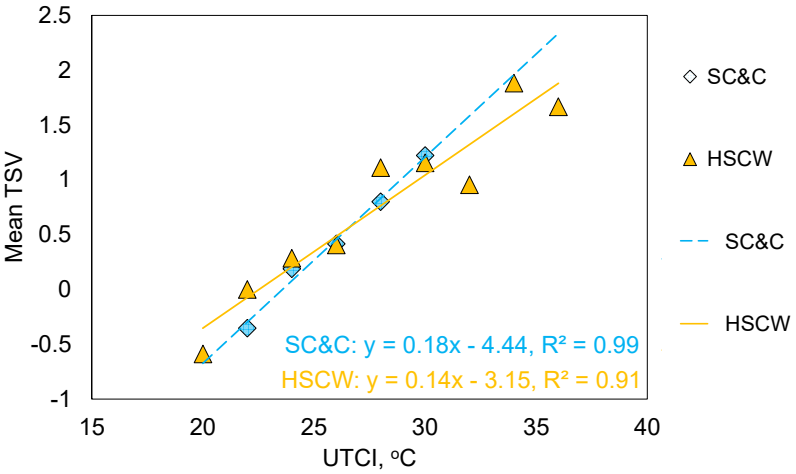


Figure 15. Relationships between UTCI and mean TSV for interviewees from severe cold (SC) and cold (C) climate regions and hot summer and cold winter (HSCW) climate regions.

4. Discussion

4.1. Comparing with Other Studies

Previous studies have provided insights into the influences of gender and age on human thermal comfort in outdoor spaces. For the effect of gender, some studies [29,47,48] have compared the distributions of the thermal sensation vote for males and females; other studies [21,24–27,49] have compared the regressed equations between the two genders. Despite the method used, all studies have shown that females were more sensitive to the thermal environment than males were, which agrees with the findings of the current study. Only Cohen et al. [28] reported that in Beer Sheva, Israel, females had shown higher tolerance to cold than males.

For the effect of age, our investigation has demonstrated that older age led to lower thermal sensation in mild and hot seasons, which agrees with the results of Lai et al. [44]. Our study has demonstrated similar sensitivity among all age groups. However, a number of studies [24,25,27,31] have shown that the elderly group was the least sensitive to the outdoor thermal environment among all age groups. It is worth noting that the data of respondents aged >50 in our study were collected under a narrow UTCI range (20–26 °C), and more data might be required to test the findings in wider conditions.

Many researchers [29,50] have discovered that, compared to foreign visitors, local residents have a higher percentage of "neutral" votes. Yang et al. [31] found that the thermal sensation of a local person was about 0.8 units higher than that of a non-local person in Umeå, Sweden. Our study further quantified the effect by showing that the overall comfort increased with longer residence time.

4.2. Clothing Behavior

The adjustment of clothing insulation is an important measure for people to adapt to the outdoor thermal environment, and many studies have documented the clothing behavior of people in their studies. Lai et al. [13] have reviewed and compared clothing behavior from different studies. This study calculates the mean clothing insulation value at different air temperatures, and the results are shown in Figure 16. The clothing insulation decreased rapidly from 21 to 25 °C, but the value became saturated thereafter as it had reached a socially acceptable threshold. In addition to the trend of the mean clothing insulation, the level of diversity (standard deviation) of the clothing insulation worn by occupants also reduced as the air temperature increased. The standard deviation trend of clothing insulation shows that adjusting clothing was a more frequently adopted measure at temperate air temperatures (within the range of our study). Figure 17 separately demonstrates the clothing insulation under different air temperatures by gender. It can be seen that females had more clothing insulation than males, especially at air temperatures between 21 and 27 °C.

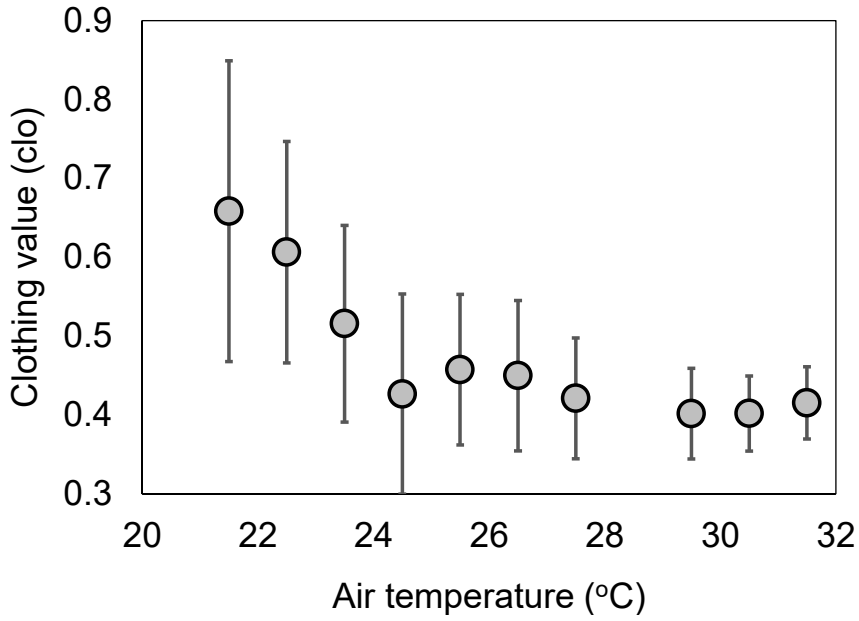


Figure 16. Clothing value under different air temperatures, with error bars indicating standard deviations.

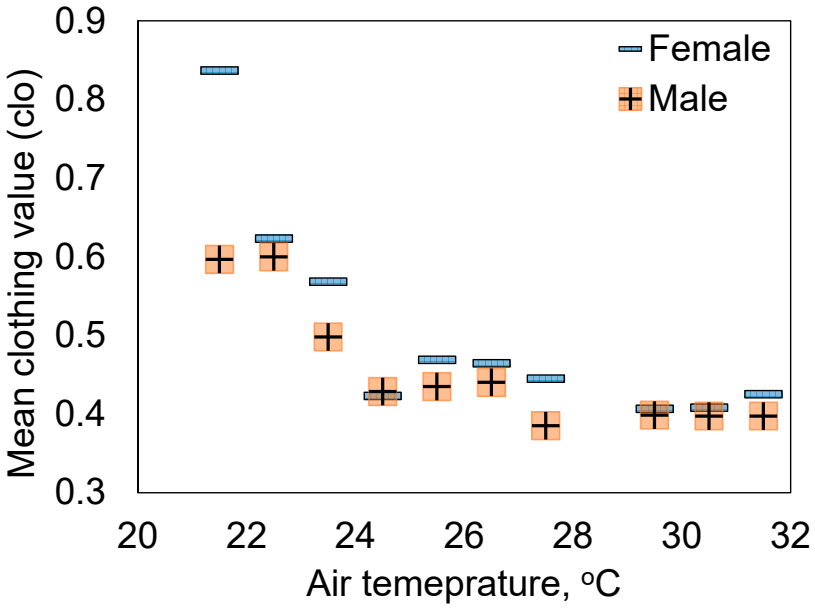


Figure 17. Clothing value under different air temperatures for males and females.

4.3. Limitations

There are several limitations of this study. The field survey was conducted in mild and hot seasons. We did not cover conditions of air temperature below 20 °C, but the winter air temperature in Shanghai can frequently be below 5 °C. As this study was only conduced in mild and hot seasons, data should also be collected in a wider thermal environment to study the impact of personal factors in various thermal environments. Accordingly, further studies should be carried out in cold seasons. Individual personal factors may amplify or diminish the influence of other factors when they interact; further analysis should be carried out on a larger dataset to study the interaction between personal factors. In addition, due to the insufficient sample size of HSWW and M climate zones, the analysis was not performed on these two climate regions to examine occupants' outdoor thermal comfort. More samples should be collected to provide enough data for further analysis, especially for the elderly age group. Thermal comfort is influenced by many factors [51–56]. Other personal factors such as personal characteristics, weight, and activity may also have an influence on outdoor thermal comfort, and these factors should be studied further.

5. Conclusions

In order to understand the outdoor thermal comfort from a gender and long-term thermal history perspective, field surveys were conducted from May to October, 2019 in outdoor spaces at a university campus in the humid subtropical climate of Shanghai, China. The field surveys obtained the microclimatic parameters and the subjective perceptions. The analysis of 520 samples has led to the following conclusions:

- The occupants generally accepted the outdoor thermal environment. During the survey, 42%, 48%, 40%, and 62% of the respondents perceived the temperature, humidity, solar radiation, and wind as neutral, respectively. In total, 57% of them felt comfortable, and only 1% and 3% stated that the overall comfort level was intolerable or very uncomfortable, respectively.
- UTCI showed a better correlation with the mean thermal sensation than PET did. A 1 K change in UTCI led to a 0.1388 unit change in mean thermal sensation.
- Female respondents demonstrated a higher sensitivity to the outdoor thermal environment than male respondents did. Different age groups had a similar level of sensitivity, while respondents younger than 20 had the highest TSV.

- Longer residence length led to a higher overall comfort rating and lower thermal sensation. Interviewees from severe cold and cold climate regions were more sensitive to the changes in UTCI than occupants from hot summer and cold winter climate regions.

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Conflicts of Interest: The authors declare no conflict of interest.

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The urban brain: Analysing outdoor physical activity with mobile EEG

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The urban brain: analysing outdoor physical activity with mobile EEG

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ABSTRACT

Background Researchers in environmental psychology, health studies and urban design are interested in the relationship between the environment, behaviour settings and emotions. In particular, happiness, or the presence of positive emotional mindsets, broadens an individual's thought-action repertoire with positive benefits to physical and intellectual activities, and to social and psychological resources. This occurs through play, exploration or similar activities. In addition, a body of restorative literature focuses on the potential benefits to emotional recovery from stress offered by green space and 'soft fascination'. However, access to the cortical correlates of emotional states of a person actively engaged within an environment has not been possible until recently. This study investigates the use of mobile electroencephalography (EEG) as a method to record and analyse the emotional experience of a group of walkers in three types of urban environment including a green space setting.

Methods Using Emotiv EPOC, a low-cost mobile EEG recorder, participants took part in a 25 min walk through three different areas of Edinburgh. The areas (of approximately equal length) were labelled zone 1 (urban shopping street), zone 2 (path through green space) and zone 3 (street in a busy commercial district). The equipment provided continuous recordings from five channels, labelled excitement (short-term), frustration, engagement, long-term excitement (or arousal) and meditation.

Results A new form of high-dimensional correlated component logistic regression analysis showed evidence of lower frustration, engagement and arousal, and higher meditation when moving into the green space zone; and higher engagement when moving out of it.

Conclusions Systematic differences in EEG recordings were found between three urban areas in line with restoration theory. This has implications for promoting urban green space as a mood-enhancing environment for walking or for other forms of physical or reflective activity.

BACKGROUND

Within environmental psychology, positive emotional mindsets[1] and specifically restorative theory[2 3] have proposed that natural settings promote recovery from stress and fatigue via attention restoration mechanisms. Soft fascination (intriguing environmental stimuli) promotes involuntary attention, enabling cognitive recovery from fatigue, and is typically present in natural settings. By contrast, hard fascination (demanding stimulation) grabs attention dramatically, increasing cognitive load, and is typically present in urban settings. The recovery from stress has been assessed in many different ways, including positive effects on heart rate,[3] perceived stress[4 5] and salivary cortisol.[6]

Brain studies are increasingly relevant as the development of affective computing[7] and brain–computer interfaces (BCI) enable researchers and designers to measure brain activity in environmental settings.[8] Although functional brain imaging has been used for subjects viewing environmental scenes in a laboratory setting,[9] no field data are available to our knowledge.

AIMS

Our study was aimed at exploring ways of using electroencephalography (EEG) technology in the assessment of urban experience.

There are two research questions.

1. Can the emotional impact of different urban environments be detected by cortical EEG signals while participants are walking through them?
2. Is there any evidence that urban green space can modify EEG signals in a restorative way, that is, reduction of stress, arousal and frustration and increase in meditation?

METHODS

Equipment

The Emotiv EPOC wireless EEG headset and accompanying software[10] was used for our research. EPOC consists of 14 sensors positioned on the wearer's scalp according to the international 10–20 system: antero-frontal (AF3, AF4, F3, F4, F7, F8), fronto-central (FC5, FC6), occipital (O1, O2), parietal (P7, P8) and temporal sites (T7, T8). Brain waves are measured in terms of amplitude (10–100 microvolts) and frequency (1–80 Hz). The four main independent bands are: δ (0.5–4 Hz), indicating deep sleep, restfulness, and conversely excitement or agitation when delta waves are suppressed; θ (4–8 Hz), indicating deep meditative states, daydreaming and automatic tasks; α (8–15 Hz), indicating relaxed alertness, restful and meditative states; β (15–30 Hz), indicating wakefulness, alertness, mental engagement and conscious processing of information. EPOC also includes a two-axis gyroscope to detect the wearer's head motion and orientation.

Emotiv EPOC is a novel, commercial device shifting from the medical EEG paradigm to affordable devices for commercial BCI use. Typical output from the device is shown in figure 1.

We do not have direct access to the manufacturer's algorithm relating to emotional parameters to raw EEG output on intellectual property rights (IPR) grounds. However, a number of studies have confirmed the reliability and validity of Emotiv's EEG performance[10 11] and the detection of emotional states.[12]

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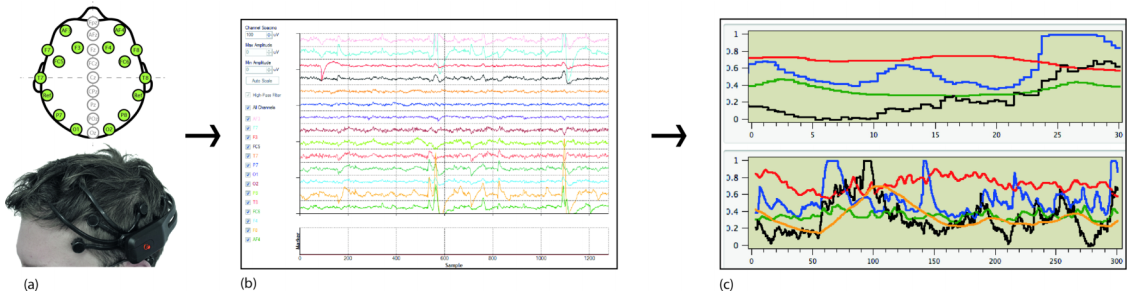


Figure 1 (A) Neural cap; (B) output from the Emotiv Testbench software: Emotiv EPOC records electroencephalography (EEG) signals from 14 sensor positions according to the 10–20 International System. Raw EEG signals are then 'translated' and classified in different emotional states and (C) output from the Emotiv Control Panel and 'Affectiv suite' (EEG data belong to the authors).

First, the EPOC EEG headset during feasibility trials[10] has been validated against a medical grade headset. The voltage and time resolution were found to be lower and the noise floor higher with the EPOC, but overall, the spectra and signals were regarded as well matched to a respected medical grade device. The amplifier linearity and channel phase differences were reported to be 'as good as any medical grade device—the resolution is limited only by the lower bit count and sampling rate'. Second, the validation of the headset has also been confirmed by a recent paper using the Emotiv mobile headgear in the *Journal of Psychophysiology*.[11] The paper concludes 'that good quality, single-trial EEG data suitable for mobile BCI can be obtained with affordable hardware'.

Our experiment involved a light, high-performing laptop with solid state storage, wireless EEG and a global positioning system (GPS) unit as shown in figure 2. For our experiments, we used Emotiv's 'Affectiv suite', which employs proprietary algorithms for filtering and translating combinations of EEG signals to four variables indicating emotional states, namely: excitement (long-term and short-term), frustration, engagement and meditation. We developed a custom-software platform to geoannotate emotional states from Emotiv's Affectiv suite in the conduct of our outdoor studies.

Participants

Twelve students from Edinburgh University were recruited to participate, and ethical scrutiny and approval for the study were provided by the School Ethics Committee. The mean age of the study participants was 30.08 with eight males, and four females.

The walk

The route was the same for all participants and it was selected to include three distinct zones in the centre of Edinburgh, of approximately equal length. These were zone 1 (urban shopping street with many people, 19th century buildings, light traffic); zone 2 (path through 26 ha green space, bordering lawns, playing fields with trees; and zone 3 (busy commercial district with heavy traffic, many pedestrians, high-noise levels). Photographs of the three zones are shown in figure 3A–C.

All of the 12 participating walkers travelled individually through the zones in the same sequence—zones 1, 2 and 3. The walkers were instructed to progress at their own pace and knew they were being observed by the researcher, who followed at a distance of 10–30 m.

All the walkers, except one, completed the walk within 24–26 min (mean 26.25 min, max 38.0 min). This generated around 48 000 lines of data (ie, around 4000 lines for each walker) across the five Emotiv channels. To reduce the volume of data, a 20% random sample was taken across all zones subsequent logistic regression analysis. This enabled two separate logistic regressions to be run with green space the common element. The consequence of this sampling resulted in

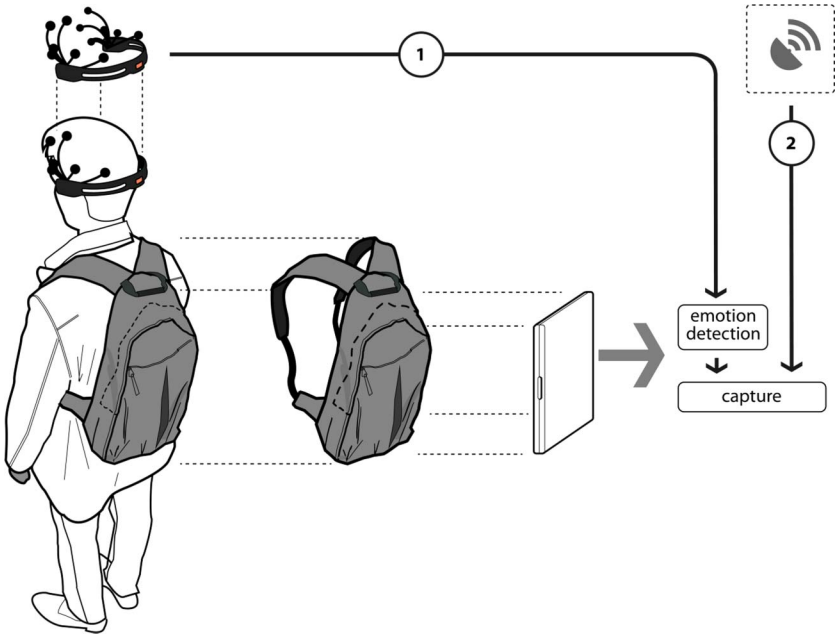


Figure 2 Diagram of the portable experimental gear. Emotiv EPOC's data (1) are geoannotated by a GPS unit and (2) stored on a laptop.

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Figure 3 (A) Zone 1—urban shopping street; (B) zone 2—green space and (C) zone 3—busy commercial district.

approximately 800 records/person corresponding to around 5 min in walk duration.

Analysis

As traditional methods of regression are unsuitable for the form of data collected, which includes multiple records per case, a note is needed on the novel analysis used. We used a form of high dimensional correlated component regression analysis with M-fold cross-validation, which has a number of benefits in addition to its capacity to deal with multiple records and multicollinearity.[13 14] First, the usually recommended limits on the number of cases per predictor are relaxed to the extent that (p), the number or predictors, can exceed (n), the number of cases—hence, its definition as *high dimensional*. Second, the method's use of regularisation prevents model overfit and delivers improved out of sample prediction in M fold cross-validation.[14]

RESULTS

At a descriptive level, a map of the route and a typical EEG record from one participant are shown in figure 4A,B. The chart at the top of figure 4B shows the emotional levels from excitement (1) and frustration (2) while graph (3) shows long-term (LT) excitement levels that have been smoothed by Emotiv's software.

A record of the fluctuations in excitement and frustration is plotted as a map according to their geographical locale in figure 4B below. For example, this participant remains excited through zones 1 and 2, which are full of social interaction, whereas excitement falls in zone 3. Conversely, frustration

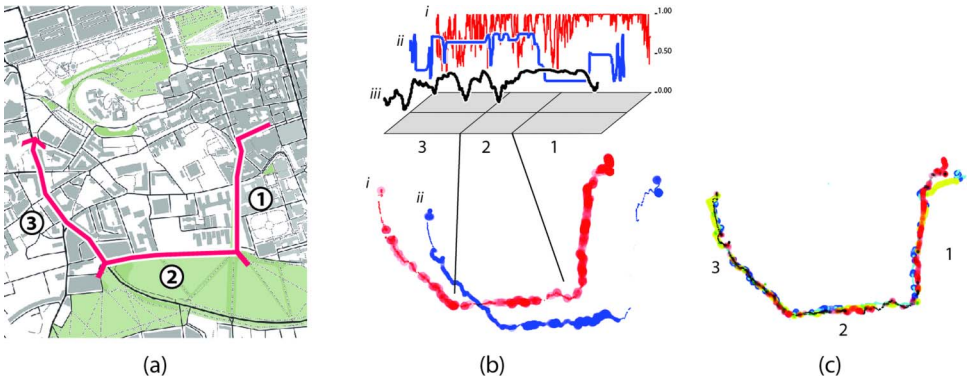


Figure 4 (A) Map of the route in central Edinburgh. (B) Emotional activity of one participant during the route, in charts (top of chart) and plot in space (bottom of chart). Red shows excitement; blue shows frustration. (C) Shows aggregate of excitement levels from the three participants. Peaks are in red, blue and yellow, respectively, for each participant.

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Table 1 The accuracy of model fit; standardised β coefficients; and frequency of each predictor in 10 rounds of 10-fold (ie, 100) regression runs

	Training	Crossvalidation	SE
Fit			
R^2	0.0592	0.0579	0.0003
AUC	0.6441	0.6423	0.0006
Accuracy	0.6694	0.6678	0.0013

Standardised coefficients

Predictors	Standardised coefficient	1.2415 CC1	0.3264 CC2	0.0742 CC3
Frustration	−0.8756	−0.8663	0.7827	−0.7496
Engagement	−0.9007	−0.6649	−0.3232	0.4077
LT excitement	−0.1829	−0.0411	−0.3663	−0.1659
Meditation	0.3812	0.2380	0.2093	0.2351

Predictor table

Predictor	All	1	2	3	4	5	6	7	8	9	10
Frustration	100	10	10	10	10	10	10	10	10	10	10
Engagement	100	10	10	10	10	10	10	10	10	10	10
LT excitement	100	10	10	10	10	10	10	10	10	10	10
Meditation	100	10	10	10	10	10	10	10	10	10	10
Total	400	40	40	40	40	40	40	40	40	40	40
Predictors		4	4	4	4	4	4	4	4	4	4

All predictors were present in all runs.
AUC, area under the curve and LT excitement, long-term excitement.

seems to be highest at the start of the walk and lowest at the end. An aggregate visualisation of the excitement levels of three participants (figure 4C—peaks are in red, blue and yellow, respectively) reveals shared patterns of emotional activity, even though the experiment was performed on different days.

Comparison of zone 1 (urban shopping street) and zone 2 (green space)

The output from correlated component logistic regression analysis is shown in table 1.

At the top of table 1 under 'Fit' is the value of R^2 from crossvalidation together with its SE. Other rows of the table show the area under the curve and accuracy of the model prediction. Note that crossvalidation on out of sample prediction is close to the training value. The effect size from R squared is 0.058, which is at a small to medium level.[15]

Below 'Fit' is a table listing the usual regression output of standardised coefficients for each of the predictors. Table 1 shows that the rank order of standardised coefficients is engagement (ie, alertness), followed by frustration, meditation and LT excitement (or longer duration arousal). The negative sign indicates that the first zone is higher on the emotional measure than the second. In other words, in moving from zones 1 to 2 (from urban street to green space), the first three predictors in the table (ie, frustration, engagement or alertness and LT excitement) became lower, whereas meditation became higher.

To the right of the standardised coefficients of predictors are three columns of the correlated components (CC1, CC2 and CC3) showing the loading of each standardised coefficient across the components. The weighting of each component is given directly above it (eg, 1.2415, 0.3264 and 0.0742). In conventional regression, the number of components K is equal to the number of predictors p (ie, K=p=4 in this case). However, regularisation, which maximises R^2 while improving out of sample prediction, produces a three component solution. The graph in figure 5A is based on this three component solution and shows the red or darker line to be maximum for R squared at four predictors.

Finally, at the bottom of table 1, the predictor table gives the output of sample prediction across 100 regression runs (10 rounds each with 10 folds) within this case of four predictors present in all 100 regression runs.

Comparison of zone 2 (green space) and zone 3 (busy commercial district)

The results for this correlated component logistic regression are shown in table 2.

In the 'fit' table, the R squared value under crossvalidation is now 0.0203, which is considered small.[15] The standard coefficient table shows that there is just one predictor, which is engagement, and only one correlated component, CC1. This is confirmed by the graph in figure 5B in which the red or darker line for one component is maximum for R squared at one predictor. In this case, the sign of the predictor is positive, indicating that engagement or alertness is higher in zone 3, the busy commercial district.

Finally, the predictor table 2 shows that for out of sample predictions across 100 regressions (10 rounds each with 10-folds), one predictor (engagement) is present across all regression runs.

DISCUSSION

This study is one of the first to use a mobile EEG system outdoors and EEG-based emotion recognition software to record emotional changes as people walk through an urban

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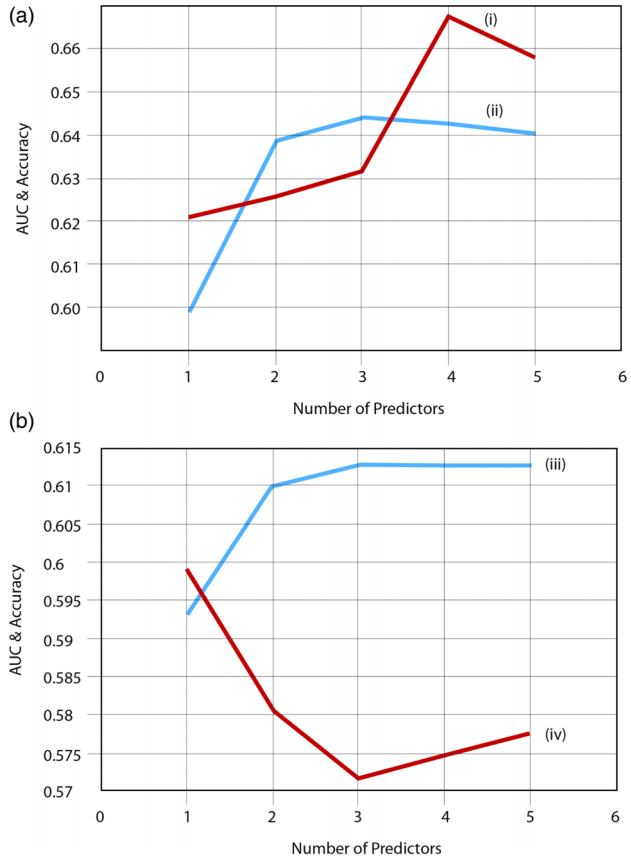


Figure 5 Graphs for area under the curve AUC (blue or light line) and cross validation accuracy (red or dark line) for the two logistic regressions. The latter is maximised at 4 predictors in (a) and 1 predictor in (b). (a) Showing the optimum four predictors for comparison zone 1 versus zone 2. (b) Showing the optimum single predictor for comparison zone 2 versus zone 3.

environment. Results confirm the research questions in showing systematic differences in processed EEG signals in different urban areas. The sequence of zones experienced was the same for all walkers. It is interesting to note that the transition from zone 1 to zone 2 (urban shopping street to green space) is in line with restorative theory with reductions in arousal, frustration, and engagement (ie, directed attention), and an increase in meditation. This change involves a greater range and subtlety of emotional response than the converse move from green space to busy commercial district (ie, heavy traffic/many people) in which one dimension of emotional change (engagement or alertness with directed attention) dominates.

In considering the limitations of the study, it would have been beneficial to have had access to the raw EEG measures, but this was not possible for the reasons given above. We also noticed that occasionally the EPOC was not streaming data, so we had to repeat the experiment or have a partial data set. In addition, owing to the ambulatory nature of the experiment, the trials produced data of varying quality with minor electrode displacements of the EPOC headset owing to the gait occasionally affecting the emotion detection algorithm.

Nonetheless, our findings are consistent with restorative theory and encourage the use of the technology to extend current research by developing novel and objective cortical correlates of emotion. This would be particularly beneficial in exploring the health improving potential of environments while people are on the move.

Contributors PA contributed to the research design and carried out data analysis. He was involved in the writing of the paper. PM contributed to the research design, recruited to the study, developed the technology, collected all data and supplied the visual images. RC contributed to the research design, interpretation of data and writing of the paper. JR inputted into the research design, supplied the restorative theory framework and the writing of the paper. We confirm that all the authors meet conditions 1, 2 and 3 below: (1) substantial contributions to the conception and design, acquisition of data, or analysis and interpretation of data; (2) drafting the article or revising it critically for important intellectual content and (3) final approval of the version to be published.

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Competing interests None.

Ethics approval The University of Edinburgh School of Arts, Culture and Environment Ethics Committee.

Provenance and peer review Not commissioned; externally peer reviewed.

Table 2 The same information for one predictor

	Training	Crossvalidation	SE
Fit			
R^2	0.0206	0.0203	0.0001
AUC	0.6119	0.5931	0.0012
Accuracy	0.5993	0.5992	0.0003

Standardised coefficients

		0.7159
Predictors	**Standardised coefficient**	**CC1**
Engagement	0.7159	1.0000

Predictor table

Predictor	All	1	2	3	4	5	6	7	8	9	10
Engagement	100	10	10	10	10	10	10	10	10	10	10
Total	100	10	10	10	10	10	10	10	10	10	10
Predictors		1	1	1	1	1	1	1	1	1	1

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The urban brain: analysing outdoor physical activity with mobile EEG

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Smart Cities (Urban Simulation, Big Data)

CityFeel - micro climate monitoring for climate mitigation and urban design

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Abstract

While a significant part of the population is concentrated in urban areas, the influence of cityscape parameters on human heat stress remain poorly understood. Yet we agree to develop urban spaces (street, square, district ...) in a way to provide best possible quality of life. In order to do so, quantitative and qualitative references are required. To fill this gap the HES-SO† - hepia/leea‡ has developed an innovative portable monitoring system that can be easily deployed in various outdoor and indoor environments. The monitoring equipment is embedded into a backpack that is carried during 'climatic urban walks' that can be reproduced at different times of the day or seasons so to yield a detailed and dynamic description of the climatic context of a portion of the city from the pedestrian point of view.

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Keywords: climate mitigation; micro climate monitoring; urban design; open data

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1. Main text

1.1. Introduction

75% of the European population is concentrated in rapidly growing urban areas. While adaptation to climate change in cities is a major challenge, "well-being" in terms of hygrothermal comfort and air quality of residents and pedestrians and the contributing effects are still poorly understood. Ongoing urban developments (streets, squares, plazas, parks, crossroads, and neighborhoods) claim to offer the best possible experience to their users (inhabitants, pedestrians). Yet, development policies are often based on subjective knowledge and lack quantitative and qualitative reference data.

Hence, climate warming and urban heat island effect have direct impact on inhabitants' health (specially the elderly and weak), on pollution levels, on energy consumption (e.g. air conditioning), on biodiversity, etc… Due to these issues, growing attention is being paid to the management of urban spaces of all kinds. Design (ergonomics, aesthetics) is not enough; climate resilient planning will be a challenge for the next decade and will help to mitigate adverse effects of climate change. Local authorities will adapt urban planning in ways that take into account climatic and sanitary constraints, meaning that urban planners and local governments require reliable reference data to make best choices with a limited budget and time.

CityFeel aims to provide a toolbox to urban planners for climate-adapted city development. This toolbox comprises a methodology, physical tools and open data that are made available to planners and the scientific community. In-depth knowledge of the dynamics of "comfort and air quality at a pedestrian level" will have many purposes, including but not limited to urban surfaces, vegetation, traffic control, health monitoring, etc.

1.2. State of the art

Unlike reference climate stations (physical parameters) and air quality stations (chemical parameters) that are commonly used for conventional meteorological purposes or the monitoring of pollution levels, *CityFeel* measures the human perception of the immediate environment in terms of response to hygrothermics and air quality. This is why the protocol must be tied to the human metabolism and measurements must take place at pedestrian level.

Most commonly, when climate or air quality data are available for a given city, these are derived from static measurement stations. Their aim is to qualify the urban climate according to conventional classifications such as city center, suburbs, countryside. For example Geneva has three WMO grade measuring stations, (maintained by the University of Geneva and by hepia / climacity) in addition to the historical station of the Swiss Office of Meteorology at Cointrin (Geneva airport). The size and complexity of these conventional stations do not allow for extensive coverage over a given territory or even a district. Furthermore, measurements from conventional weather stations do not describe what is "felt" at pedestrian level.

Other methods consist of making use of numerical simulation models; whilst the complexity of these algorithms can be managed with increase of computing power, limitations are mainly due to the capability of physical model input description and limited resolution of the city model, as well as the scarcity of reliable data for model calibration. These models tend to have limited resolution and reliability at pedestrian level, where the feedback is most critical and also is most complex to simulate.

Drive- or walkthroughs with sensors attached to vehicles have been demonstrated over the past decades and have produced parameter profiles trough many city sections. The observation that for example 'green' or 'blue' do not always have measurable effect shows that there still is a lack of knowledge on how these indicators should be interpreted into robust 'climatic bricks' for urban developers and planners.

Due to the previous points we have not yet found a city with a network of long-term quality climate measurements at pedestrian level with a high density of measurement points and who's monitoring considers human metabolism.

1.3. Methodology

The first step was to define an investigation methodology and the qualitative criteria based on the multiple senses that affect the complex and multisensory perception of pedestrians while roaming through the city (e.g. the human body is sensitive to radiant temperature from surrounding surfaces, to air turbulence, to noise, to odors, to visual and acoustic perception, and so on…).

Then, knowing that the deployment of a fixed monitoring network with as many data collection points as necessary would be too restrictive, too expensive and difficult to achieve, we changed the paradigm by developing a portable and affordable measuring device in the form of a backpack. In this way, the sensors are attached to a pedestrian operator and move along a free or predefined path through the city. This provides great flexibility of use, anywhere and at any given time. The data is collected at a rate of 1 Hz and is made available for display and analysis by the means of a dedicated graphical interface or more traditional analysis tools.

Hemispherical and ground images complete the sensor array. They are used for analysis, such as sky view factor or vegetation index, but also allow for 'reverse engineering' of acquired data: in a way they give eyes to the sensors and allow better understanding of cause and effect. Data collected from the sensors feed a human body heat balance model to evaluate physiological equivalent temperature (PET), which is defined as equivalent temperature having comparable heat sensation in controlled environmental conditions without radiation, wind, etc. PET is easily understandable and is closest to the actual heat sensation.



Fig. 1. (left) operator during a climate walk, carrying the microclimate-meter, (right) sample of data displayed in the ad-hoc viewer

2. Description

2.1. Investigation and measurement equipment

The specifically designed 'microclimate-meter', further referred to as *uCM*, is the core part of *CityFeel*. The goal is to deploy an array of environmental sensors within a mobile and light-weight system.

Thus, beyond the conventional parameters (ambient temperature and humidity), the *uCM* also logs multi-directional radiation temperatures, vertical temperature profiles at pedestrian level, thermal turbulence, wind speed, direction and turbulence, noise level and records the soundtrack and hemispherical images (sky and ground camera). A GPS receiver and an inertial reference platform make it possible to locate the measured data in time and in space. In the latest version we added additional sensors to measure air quality such as CO2, NOx, Ozone and micro particles.

Table 1. Table of physical sensors:

Sensor	Resolution	Absolute error
Ambient temperature	0.1°C	0.2°C
High speed temperature	0.1°C	0.4°C
Relative humidity	0.1%	1%
Radiant temperature (left, right … up, down)	0.1°C	0.5°C
Solar radiation	1W/m2	1.5%
Wind speed	0.01m/s	1%
Sound pressure	1dB	3%
Position (GPS)	0.1m	<10m

2.2. The uCM on-board equipment

The sensor array is connected to a dedicated logging equipment consisting of a Campbell Scientific CR-6 data logger, an ATmega328 based micro-controller board and a nano-PC for image handling and on-the-fly data quality assessment and synchronization. Ongoing development includes a real-time graphical operator interface.

The approximate data flux is 7MB/second. The autonomy is about three hours of continuous sensor streaming, limited only to LiPo battery and memory stick capacity and capacity of the operator to carry the 5kg equipment.

2.3. Data processing and results

During the development process, calibration test walks followed a fixed reference sequence, characterized by a sufficiently contrasting "urban scape" (avenue, square, park, street). At present, the operator simply puts on the *uCM* and starts monitoring according to specific interests.

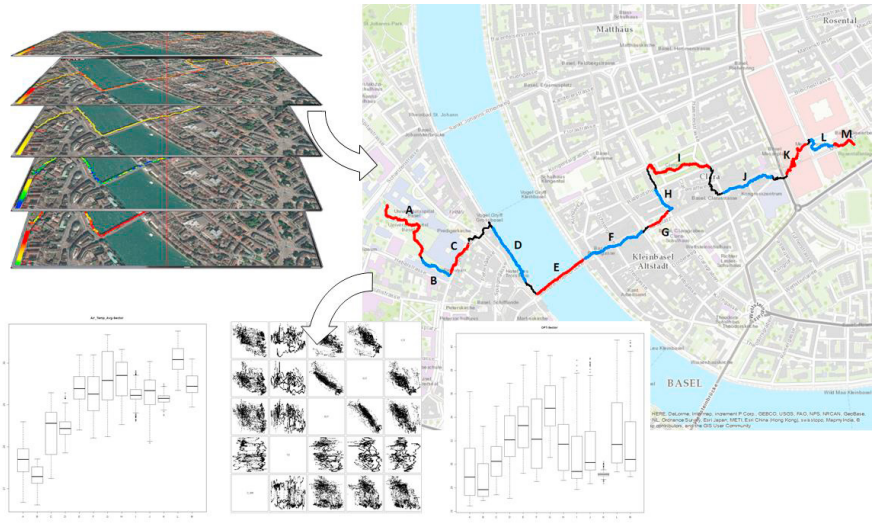


Fig. 2. Overview of the process of data analysis: crude data visualization, sectorization, statistical analysis

Once the campaign is finished, the operator transfers the collected data to a desktop PC where a script aggregates all collected data, converts it to a suitable format for post-treatment and calculates derived indicators related to outdoor human heat stress, such as mean radiant and operative temperatures, PET (Physiologically Equivalent Temperature), UTCI (Universal Thermal Climate Index), HUMIDEX (Humidity index), WBGT (Wet Bulb Globe Temperature), etc.

A Java™ program allows for first observation and interpretation of the collected data by the means of a dedicated graphical user interface. In depth analysis is done by traditional environments, such as e.g. ArcGIS mapping and

analytics software, Mathlab®, Statistica® or R, typically used to sectorise, analyze and visualize data. Machine learning algorithms take care of related image analysis, such as sky view factor (SVF) or vegetation indexes.

2.4. Case-studies

The interaction of city and climate is a very complex process and is determined by many influencing factors. Added to this are weather conditions, time of day, use (differences between works and holidays) and much more. Never less, assuming that variations over a short period of time are small, a close look into measurements data give clues to comprehension of the urban metabolism.

How efficient are climate strategies in the real-world and what are the best options can be identified by the observation of comparative solutions at different periods and times of the day. For example evaporation from water effectively acts as a cooling strategy, as can be seen on the illustration hereafter.

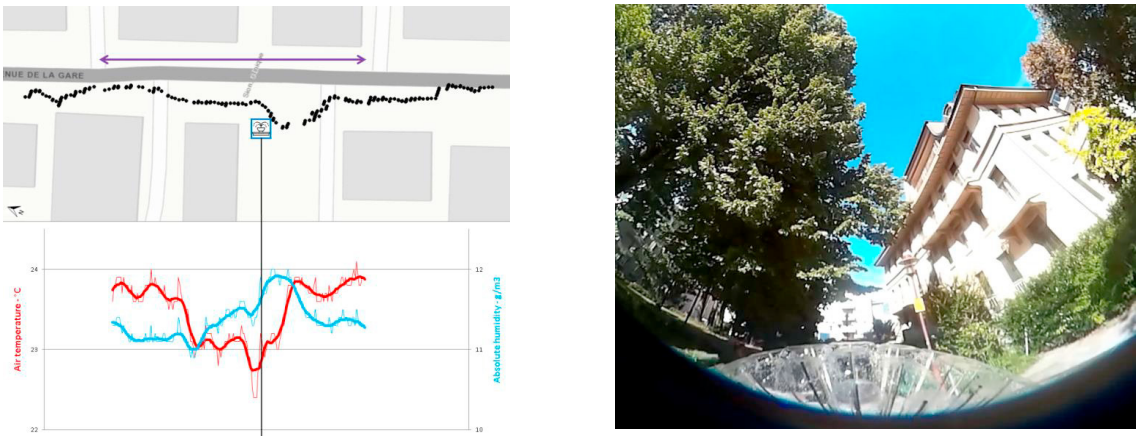


Fig. 3. Example effect of water as urban climatic design feature decreasing temperature whilst increasing air water content.

Climate walks are also useful to identify places where heat stress and pollution are least. This knowledge can be used to inform the population and to identify local adaptation and mitigation strategies. Charts showing parameters such as PET, UTCI, humidity, wind, etc… show a strong correlation between urban morphology and parameters affecting human outdoor comfort and health.

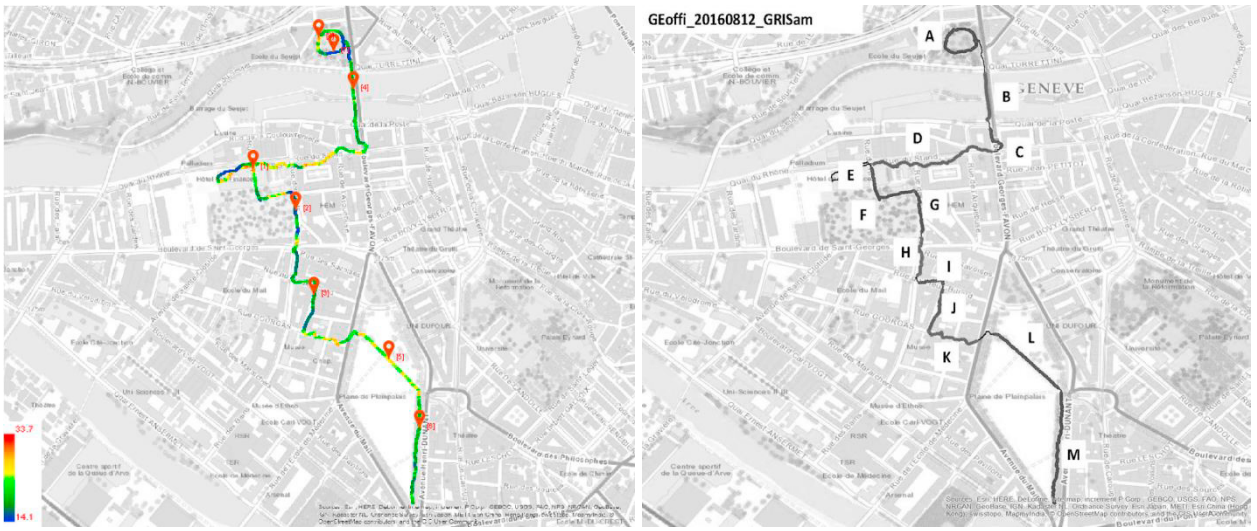


Fig. 4. PET profile of a climate walk in Geneva center indicating cold- and hot spots on a summer morning and sectors (right)

It has been shown that already a single climate walk can be meaningful. The following comparison of trends for a series of thirteen sectors (A to M, see figure 4 right side) on different days is very good. This means that even though the absolute temperature may not be known in advance, the data of a single walk can be extrapolated to predict thermal behavior of the city and give objective indications to questions such as: what are the best places to rest, which are the most comfortable and healthiest ways to travel from one place to another depending on the time of day, and so on.

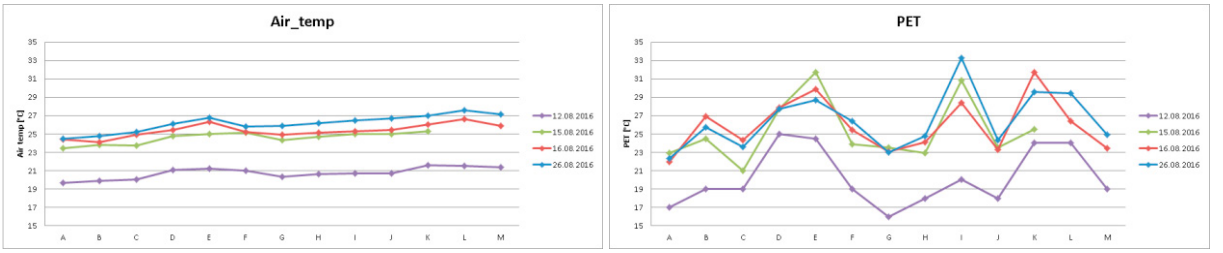


Fig. 5. Comparison of temperature and PET profiles for different days; sectors refer to figure 4

2.5. Discussion and outlook

The *uCM* monitoring and logging backpack has contributed in research projects related to effects of vegetation and water in the city and urban morphology. Numerous climatic walkthroughs have been completed over the last year revealing interesting insight into climatic metabolism of cities.

The *uCM* is readily available for research in the field of city climate, city- and soundscapes (well-being) and air quality (health). On the basis of 'crowdsharing' the *uCM* is made available for free to public and academic partners who are interested in investigating, contributing and sharing studies through an open-data structure containing monitored situations relating to climatic urban design of public spaces and air quality. The collected data will be shared amongst the scientific community by the means of an open data structure. This knowledge will be useful in applications such as urban planning, choice of urban surfaces, design of the "green city" and uses such as traffic control, pollution prediction, health assessment as well as general urban management.

The *uCM* is now in version 2.0 and ongoing hard- and software development aim to make the *uCM* ever more immersive, easier and intuitive to use.

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